

Category 2: Algorithm-Level DP Questions

All Questions with Metadata

Total: 125 questions

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1. Accounting and Composition

1.1. Accounting

Question 1

Question:

Let $\mathcal{M}$ be an ε -DP Laplace mechanism. For any $\alpha > 1$, does $\mathcal{M}$ satisfy (α, ε') -RDP with

$$\varepsilon'(\alpha) = \frac{1}{\alpha - 1} \log \left(\frac{\alpha}{2\alpha - 1} e^{(\alpha-1)\varepsilon} + \frac{\alpha - 1}{2\alpha - 1} e^{-\alpha\varepsilon} \right)$$

.

Label	Yes (True)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	atom
Related Question	1
Citation	2130
Section	VI
Publish Year	2017
PDF Link	https://arxiv.org/pdf/1702.07476 [Mironov, 2017]
Comments	The bound is stated in Mironov [2017, Proposition 6].

Question 2

Question:

Let $\mathcal{M}$ be an ε -DP Laplace mechanism. For any $\alpha > 1$, does $\mathcal{M}$ satisfy (α, ε') -RDP with

$$\varepsilon'(\alpha) = \frac{1}{\alpha} \log \left(\frac{\alpha}{2\alpha - 1} e^{(\alpha-1)\varepsilon} + \frac{\alpha - 1}{2\alpha - 1} e^{-\alpha\varepsilon} \right).$$

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	conclusion_perturbation
Related Question	1
Citation	2130
Section	VI
Publish Year	2017

PDF Link	https://arxiv.org/pdf/1702.07476 [Mironov, 2017]
Comments	The RDP bound for the Laplace mechanism given in Mironov [2017, Proposition 6] is tight. Since the claimed statement asserts a strictly smaller upper bound via replacing factor $\frac{1}{\alpha-1}$ with $\frac{1}{\alpha}$, it cannot hold in general.

Question 3

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be a collection of mechanisms, each satisfying ε -Differential Privacy. Does the sequential composition of these mechanisms satisfy $(2\sqrt{n}\Phi^{-1}(e^\varepsilon/(e^\varepsilon + 1)))$ -Gaussian Differential Privacy (GDP)? (Φ denotes the cumulative distribution function (CDF) of the standard normal distribution)

Label	Yes (True)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	atom
Related Question	3
Citation	6
Section	lemma 15
Publish Year	2024
PDF Link	https://arxiv.org/pdf/2210.09269 [Zhu et al., 2022]
Comments	By Liu et al. [2022, Theorem 5.1], ε -DP implies $2\Phi^{-1}(e^\varepsilon/(e^\varepsilon + 1))$ -GDP, by composition of GDP (Dong et al. [2022, Corollary 2]), the composed mechanism satisfies $2\sqrt{n}\Phi^{-1}(e^\varepsilon/(e^\varepsilon + 1))$ -GDP.

Question 4

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be a collection of mechanisms, each satisfying ε -Differential Privacy. Does the sequential composition of these mechanisms satisfy $(\sqrt{n}\Phi^{-1}(e^\varepsilon/(e^\varepsilon + 1)))$ -Gaussian Differential Privacy (GDP)? (Φ denotes the cumulative distribution function (CDF) of the standard normal distribution)

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	conclusion_perturbation
Related Question	3
Citation	6
Section	lemma 15
Publish Year	2024

PDF Link <https://arxiv.org/pdf/2310.14661> [Lin et al., 2023]

Comments Consider the following counter example. Let $n = 1$ and consider random response mechanism $\mathcal{M}$ with $p = \frac{\exp(\varepsilon)}{1+\exp(\varepsilon)}$. Then $\mathcal{M}$ satisfies ε -DP and we can directly calculate the TV distance $dist_1 := \text{TV}(\mathcal{M}(x), \mathcal{M}(x')) = \frac{\exp(\varepsilon)-1}{\exp(\varepsilon)+1}$. Notice that $\text{TV}(\mathcal{N}(0, 1), \mathcal{N}(\mu, 1)) = 2\Phi(\mu/2) - 1 := dist_2$. Let $\mu = \Phi^{-1}\left(\frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}\right)$ and set $\varepsilon = 1$, we have $dist_1 \approx 0.4621$ and $dist_2 \approx 0.2419$, this implies $(\mathcal{M}(X), \mathcal{M}(X'))$ is more distinguishable than $(\mathcal{N}(0, 1), \mathcal{N}(\mu, 1))$, so $\mathcal{M}$ doesn't satisfy μ -GDP. Therefore, the statement does not hold in general.

Question 5

Question:

Let $\mathcal{M}$ satisfy ε -pure Gaussian Differential Privacy. Does $\mathcal{M}$ also satisfy $f(\varepsilon)$ -DP? Where $f(x) = \log(\frac{\Phi(\varepsilon/2)}{1-\Phi(\varepsilon/2)})$ and Φ denotes the cumulative distribution function of the standard normal distribution.

Label No (False)

Subject Accounting and Composition

Topic Accounting

Negative Mode atom

Related Question 5

Citation 736

Section None

Publish Year 2022

PDF Link <https://arxiv.org/abs/1905.02383> [Dong et al., 2022]

Comments Let $\mathcal{M}$ be a ε -pure GDP gaussian mechanism, then $\mathcal{M}$ doesn't satisfy pure DP.

Question 6

Question:

Let $\mathcal{M}$ be an exponential mechanism satisfy ε -DP. Does $\mathcal{M}$ also satisfy $\frac{\varepsilon^2}{8}$ -zero concentrated DP?

Label Yes (True)

Subject Accounting and Composition

Topic Accounting

Negative Mode atom

Related Question 6

Citation 45

Section	3
Publish Year	2020
PDF Link	https://arxiv.org/abs/2004.07223 [Cesar and Rogers, 2020]
Comments	The exponential mechanism satisfies the bounded range property [Durfee and Rogers, 2019, Dong et al., 2019]. Using this improved analysis, one can further show that it satisfies $\frac{\epsilon^2}{8}$ -zCDP.

Question 7

Question:

Let $\mathcal{M}$ be a Permute-and-Flip mechanism satisfy ϵ -DP. Does $\mathcal{M}$ also satisfy $\frac{\epsilon^2}{8}$ -zero concentrated DP?

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	assumption_perturbation
Related Question	6
Citation	26
Section	None
Publish Year	2021
PDF Link	https://arxiv.org/abs/2105.07260 [Ding et al., 2021]
Comments	Permute-and-flip [McKenna and Sheldon, 2020] does not satisfy the bounded-range condition [DifferentialPrivacy.org, 2020]. Consequently, under the standard conversion, it guarantees only $\epsilon^2/2$ -DP and does not achieve $\epsilon^2/8$ -zCDP.

Question 8

Question:

Let $\mathcal{M}$ be an ϵ -DP Laplace mechanism. For any $\alpha > 1$, does $\mathcal{M}$ satisfy (α, ϵ') -RDP with $\epsilon' = \epsilon - \frac{1}{\alpha-1} \log\left(\frac{1+e^{-\epsilon}}{1+e^{-(2\alpha-1)\epsilon}}\right)$

Label	Yes (True)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	atom
Related Question	8
Citation	None
Section	None

Publish Year	2024
PDF Link	https://differentialprivacy.org/pdp-to-zcdp/ [DifferentialPrivacy.org, 2021]
Comments	It follows by DifferentialPrivacy.org [2021, Theorem 4]. We note that DifferentialPrivacy.org [2021, Theorem 4] holds for any ε -DP mechanism, thus it also holds for ε -DP Laplace Mechanism.

Question 9

Question:

Let $\mathcal{M}$ be an ε -DP Laplace mechanism. For any $\alpha > 1$, does $\mathcal{M}$ satisfy (α, ε') -RDP with $\varepsilon' = \varepsilon - \frac{2}{\alpha-1} \log\left(\frac{1+e^{-\varepsilon}}{1+e^{-(2\alpha-1)\varepsilon}}\right)$

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	conclusion_perturbation
Related Question	8
Citation	None
Section	None
Publish Year	2024
PDF Link	https://differentialprivacy.org/pdp-to-zcdp/ [DifferentialPrivacy.org, 2021]
Comments	Notice that ε' is not a valid RDP upper bound. For example, taking $\alpha = 2$ and $\varepsilon = 0.1$ yields $\varepsilon' \approx -0.7 < 0$ which is impossible since Rényi divergence must be nonnegative.

Question 10

Question:

Let $\mathcal{M}$ be an (ε, δ) -DP Gaussian mechanism. For any $\alpha > 1$, does $\mathcal{M}$ satisfy (α, ε') -RDP with $\varepsilon' = \varepsilon - \frac{1}{\alpha-1} \log\left(\frac{1+e^{-\varepsilon}}{1+e^{-(2\alpha-1)\varepsilon}}\right) + \delta$

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	atom
Related Question	10
Citation	None
Section	None
Publish Year	2024

PDF Link	https://differentialprivacy.org/pdp-to-zcdp/ [DifferentialPrivacy.org, 2021]
Comments	As $\alpha \rightarrow \infty$, we have $\varepsilon' \rightarrow \varepsilon + \delta$, which would imply that $\mathcal{M}$ is $(\varepsilon + \delta)$ -DP. However, the Gaussian mechanism does not satisfy pure DP for any finite ε . Therefore, the statement cannot be true.

Question 11

Question:

Let $\mathcal{M}$ be an ε -DP mechanism. Does $\mathcal{M}$ satisfy $\frac{e^\varepsilon - 1}{e^\varepsilon + 1}\varepsilon$ -zero concentrated DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	atom
Related Question	11
Citation	None
Section	None
Publish Year	2024
PDF Link	https://differentialprivacy.org/pdp-to-zcdp/ [DifferentialPrivacy.org, 2021]
Comments	The statement follows DifferentialPrivacy.org [2021, Theorem 5].

Question 12

Question:

Let $\mathcal{M}$ be an ε -DP Laplace mechanism. Does $\mathcal{M}$ satisfy $\frac{e^\varepsilon - 1}{e^\varepsilon + 2}\varepsilon$ -zero concentrated DP?

Label	No (False)
Subject	Accounting and Composition
Topic	Accounting
Negative Mode	conclusion_perturbation
Related Question	12
Citation	None
Section	None
Publish Year	2024
PDF Link	https://differentialprivacy.org/pdp-to-zcdp/ [DifferentialPrivacy.org, 2021]
Comments	By Harrison and Manurangsi [2025, Theorem 5], the tight zCDP bound for the ε -DP Laplace mechanism is $\varepsilon + e^{-\varepsilon} - 1$. Taking $\varepsilon = 1$, we obtain

$\frac{e^\varepsilon - 1}{e^\varepsilon + 2} \varepsilon \Big|_{\varepsilon=1} \approx 0.3642$ and $(\varepsilon + e^{-\varepsilon} - 1) \Big|_{\varepsilon=1} \approx 0.3679$. Since the proposed value is smaller than the tight bound, the statement cannot be true.

1.2. Amplification by Subsampling

Question 13

Question:

Let $\mathcal{M}$ satisfy (ε, δ) -DP under the add/remove neighbouring relationship. Consider Poisson subsampling with inclusion rate γ , i.e., each individual is included independently with probability γ . Does the subsampled mechanism satisfies $(\log(1 + \gamma(e^\varepsilon - 1)), \gamma\delta)$ -DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Amplification by Subsampling
Negative Mode	atom
Related Question	13
Citation	513
Section	4
Publish Year	2018
PDF Link	https://arxiv.org/pdf/1807.01647 [Balle et al., 2018]
Comments	This statement follows Balle et al. [2018, Theorem 8].

Question 14

Question:

Let $\mathcal{M}$ satisfy (ε, δ) -DP under the add/remove neighbouring relationship. Consider Poisson subsampling with inclusion rate γ , i.e., each individual is included independently with probability γ . Does the subsampled mechanism satisfies $(\log(1 - \gamma e^\varepsilon - \gamma), \gamma\delta)$ -DP?

Label	No (False)
Subject	Accounting and Composition
Topic	Amplification by Subsampling
Negative Mode	conclusion_perturbation
Related Question	13
Citation	513
Section	4
Publish Year	2018
PDF Link	https://arxiv.org/pdf/1807.01647 [Balle et al., 2018]

Comments By Balle et al. [2018, Theorem 12], Balle et al. [2018, Theorem 8] is tight. The statement is strictly stronger than the one in Balle et al. [2018, Theorem 8], and therefore does not hold in general.

Question 15

Question:

Let $\mathcal{M}$ satisfy (ε, δ) -DP under the add/remove neighbouring relationship. Consider sampling without replacement, where a subset of size m is drawn uniformly at random from a dataset of size n . Does the resulting mechanism satisfies $\left(\log\left(1 + \frac{m}{n}(e^\varepsilon - 1)\right), \frac{m}{n}\delta\right)$ -DP?

Label No (False)

Subject Accounting and Composition

Topic Amplification by Subsampling

Negative Mode assumption_perturbation; algorithm_perturbation

Related Question 17

Citation 513

Section 4

Publish Year 2018

PDF Link <https://arxiv.org/pdf/1807.01647> [Balle et al., 2018]

Comments Privacy amplifications through sampling without replacement does not support the add/remove neighboring since sample size n is private information.

Question 16

Question:

Let $\mathcal{M}$ satisfy (ε, δ) -DP under the replace-one neighbouring relationship. Consider sampling without replacement, where a subset of size m is drawn uniformly at random from a dataset of size n . Does the resulting mechanism satisfies $\left(\log\left(1 + \frac{m}{2n}(e^\varepsilon - 1)\right), \frac{m}{n}\delta\right)$ -DP?

Label No (False)

Subject Accounting and Composition

Topic Amplification by Subsampling

Negative Mode conclusion_perturbation

Related Question 17

Citation 513

Section 4

Publish Year 2018

PDF Link <https://arxiv.org/pdf/1807.01647> [Balle et al., 2018]

Comments By Balle et al. [2018, Theorem 12], Balle et al. [2018, Theorem 9] is tight. The bound stated here is smaller than that of Balle et al. [2018, Theorem 9] by a factor of 2, and therefore does not hold in general.

Question 17

Question:

Let $\mathcal{M}$ satisfy (ε, δ) -DP under the replace-one neighbouring relationship. Consider sampling without replacement, where a subset of size m is drawn uniformly at random from a dataset of size n . Does the resulting mechanism satisfies $\left(\log\left(1 + \frac{m}{n}(e^\varepsilon - 1)\right), \frac{m}{n}\delta\right)$ -DP?

Label Yes (True)

Subject Accounting and Composition

Topic Amplification by Subsampling

Negative Mode atom

Related Question 17

Citation 513

Section 4

Publish Year 2018

PDF Link <https://arxiv.org/pdf/1807.01647> [Balle et al., 2018]

Comments By Balle et al. [2018, Theorem 9].

1.3. Parallel Composition

Question 94

Question:

We aim to release a histogram whose bins are n disjoint, data-independent intervals. If we add independent noise drawn from $\text{Lap}(2/\varepsilon)$ to each bin, does the resulting mechanism satisfy ε -differential privacy under the replace-one neighboring relationship?

Label Yes (True)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode atom

Related Question 94

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments The sensitivity for releasing the histogram under the replace-one neighboring relation is 2. Therefore, adding independent Laplace noise with scale $\varepsilon/2$ to each histogram bin guarantees ε -differential privacy.

Question 95

Question:

We aim to release a histogram whose bins are n data-independent but potentially non-disjoint intervals. If we add independent noise drawn from $\text{Lap}(2/\varepsilon)$ to each bin, does the resulting mechanism satisfy ε -differential privacy under the replace-one neighboring relationship?

Label No (False)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode assumption_perturbation

Related Question 94

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments Since the histogram bins may be non-disjoint, replacing a single data point can affect the counts of more than two bins. Consequently, adding Laplace noise with scale $2/\varepsilon$ may not be sufficient to guarantee ε -differential privacy.

Question 96

Question:

We aim to release a histogram whose bins are n disjoint, data-independent intervals. If we add independent noise drawn from $\text{Lap}(1/\varepsilon)$ to each bin, does the resulting mechanism satisfy ε -differential privacy under the replace-one neighboring relationship?

Label No (False)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode algorithm_perturbation

Related Question 94

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments The sensitivity for releasing the histogram under the replace-one neighboring relation is 2. Therefore, adding independent Laplace noise with scale ε to each histogram bin only guarantees 2ε -differential privacy.

Question 97

Question:

We aim to release a histogram whose bins are n disjoint, data-independent intervals. If we add independent noise drawn from $\text{Lap}(1/\varepsilon)$ to each bin, does the resulting mechanism satisfy ε -differential privacy under the add/remove neighboring relationship?

Label Yes (True)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode conclusion_perturbation

Related Question 94

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments The sensitivity for releasing the histogram under the add/remove neighboring relation is 1. Therefore, adding independent Laplace noise with scale ε to each histogram bin guarantees ε -differential privacy.

Question 98

Question:

We aim to release a histogram whose bins are n disjoint, data-dependent intervals. If we add independent noise drawn from $\text{Lap}(1/\varepsilon)$ to each bin, does the resulting mechanism satisfy ε -differential privacy under the add/remove neighboring relationship?

Label No (False)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode assumption_perturbation; conclusion_perturbation

Related Question 94

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments Since the histogram bins are data-dependent, add/remove a single data point can potentially affect the counts of more than one bins. Consequently, adding Laplace noise with scale ε may not be sufficient to guarantee ε -differential privacy.

Question 99

Question:

Consider the following mechanism: $\mathcal{M}(X) := \{\mathcal{M}_i(X_i)\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy?

Label Yes (True)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode atom

Related Question 99

Citation 1997

Section 2.4.2

Publish Year 2009

PDF Link <https://dl.acm.org/doi/10.1145/1810891.1810916> [McSherry, 2010]

Comments Since the histogram bins are data-dependent, add/remove a single data point can potentially affect the counts of more than one bins. Consequently, adding Laplace noise with scale ε may not be sufficient to guarantee ε -differential privacy.

Question 100

Question:

Consider the following mechanism: $\mathcal{M}(X) := \{\mathcal{M}_i(X_{n-i+1})\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy?

Label Yes (True)

Subject Accounting and Composition

Topic Parallel Composition

Negative Mode algorithm_perturbation

Related Question 99

Citation 1997

Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Under the assumptions that subsets are disjoint and the partition is data-independent, the combined mechanism satisfies parallel composition, thus $\mathcal{M}$ satisfies $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy.

Question 101

Question:

Consider the procedure that releases $\mathcal{M} := \{a_i\}_{i=1}^n$. The outputs are defined recursively by $a_1 = \mathcal{M}_1(X_1)$, $a_2 = (a_1, \mathcal{M}_2(a_1, X_2))$, $a_3 = (a_1, a_2, \mathcal{M}_3(a_1, a_2, X_3))$, and in general $a_n = (a_1, \dots, a_{n-1}, \mathcal{M}_n(a_1, \dots, a_{n-1}, X_n))$. Assume the datasets are disjoint and cover the whole data, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^n X_i = X$. Suppose each mechanism $\mathcal{M}_i$ is $(\varepsilon_i, \delta_i)$ -DP with respect to its own subset X_i . Furthermore, the partition is data-independent. Does the overall release $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	atom
Related Question	101
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Under the assumptions that the subsets are disjoint and the partition is data-independent, the combined mechanism follows the parallel composition and post-processing property of differential privacy, this implies that $\mathcal{M}$ satisfies $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy.

Question 102

Question:

Consider releasing $\mathcal{M} := \{a_i\}_{i=1}^n$, where $a_1 = \mathcal{M}_1(X_1)$, $a_2 = \mathcal{M}_2(a_1, X_2)$, $a_3 = \mathcal{M}_3(a_1, a_2, X_3)$, and in general $a_i = \mathcal{M}_i(a_1, \dots, a_{i-1}, X_i)$ for $i \geq 2$. Assume the data partitions are disjoint and cover the full dataset, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$ and $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ is $(\varepsilon_i, \delta_i)$ -DP with respect to its own subset X_i . Does the overall release $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Parallel Composition

Negative Mode	algorithm_perturbation
Related Question	101
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Under the assumptions that the subsets are disjoint and the partition is data-independent, the combined mechanism follows the parallel composition and post-processing property of differential privacy, this implies that $\mathcal{M}$ satisfies $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy.

Question 103

Question:

Consider this procedure: release $\mathcal{M} := \{a_i\}_{i=1}^n$ with $a_1 = \mathcal{M}_1(X_1)$, $a_2 = (\mathcal{M}_1(X_1), \mathcal{M}_2(\mathcal{M}_1(X_1), X_2))$, $a_3 = (\mathcal{M}_1(X_1), \mathcal{M}_2(\mathcal{M}_1(X_1), X_2), \mathcal{M}_3(\mathcal{M}_1(X_1), \mathcal{M}_2(\mathcal{M}_1(X_1), X_2), X_3))$ and all other a_i 's are defined similarly. Here $\mathcal{M}_i(X_j)$ means apply the mechanism $\mathcal{M}_i$ on subset X_j . Suppose $X_i \cap X_j = \emptyset$ for any $i \neq j$ and $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -DP. Does $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -DP?

Label	No (False)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	algorithm_perturbation
Related Question	101
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Under the assumptions that the subsets are disjoint and the partition is data-independent, the combined mechanism follows the parallel composition and post-processing property of differential privacy, this implies that $\mathcal{M}$ satisfies $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy.

Question 104

Question:

Consider the following procedure: $\mathcal{M} := \{\mathcal{M}_i(X_i)\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\varepsilon_i, \delta_i)$ -differential privacy?

Label	No (False)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	conclusion_perturbation
Related Question	99
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Under the assumptions that the subsets are disjoint and the partition is data-independent, the combined mechanism follows the parallel composition and post-processing property of differential privacy, this implies that $\mathcal{M}$ satisfies $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy, rather than the stronger $(\varepsilon_i, \delta_i)$ -DP guarantee stated.

Question 105

Question:

Consider the following mechanism: $\mathcal{M}(X) := \{\mathcal{M}_i(X_{i \bmod 2})\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy?

Label	No (False)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	algorithm_perturbation
Related Question	99
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	The overall mechanism can be written as $\mathcal{M} := \left\{ \{\mathcal{M}_i(X_0)\}_{i \in A}, \{\mathcal{M}_i(X_1)\}_{i \in B} \right\}$, where $A := \{i \in [n] : i \text{ is even}\}$ and $B := \{i \in [n] : i \text{ is odd}\}$. Consequently, $\mathcal{M}$ satisfies $\left(\max \left\{ \sum_{i \in A} \varepsilon_i, \sum_{i \in B} \varepsilon_i \right\}, \max \left\{ \sum_{i \in A} \delta_i, \sum_{i \in B} \delta_i \right\} \right)$ -DP

Question 106

Question:

Consider the following procedure: $\mathcal{M} := \{\mathcal{M}_i(X_i)\}_{i=1}^n$, where $X_i \cap X_{i+1} = \emptyset$ for all $i \in \{1, \dots, n-1\}$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy?

Label	No (False)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	assumption_perturbation
Related Question	99
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	Consider the following counter example: let $X_i = X$ when i is odd and $X_i = \emptyset$ when i is even, then $\mathcal{M}$ satisfies $\mathcal{M}$ satisfies $\left(\sum_{i \in [n] \text{ and } i \text{ is odd}} \varepsilon_i, \sum_{i \in [n] \text{ and } i \text{ is odd}} \delta_i \right)$ -DP

Question 107

Question:

Given dataset $X = \{x_1, \dots, x_n\} \subset \mathbb{R}$, define empirical cumulative distribution function $F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{x_i \leq x\}} = \frac{\#\{i | x_i \leq x\}}{n}$ and its inverse: $F_n^{-1}(p) = \inf \{x \in \mathbb{R} \mid F_n(x) \geq p\}$, for $p \in [0, 1]$. Define subset of X as $X_i := \{x \in X \mid F_n^{-1}(\frac{i-1}{n}) < x \leq F_n^{-1}(\frac{i}{n})\}$, for $i \in \{1, \dots, n\}$. Clearly each subset is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$ -differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$ -differential privacy under replace-one neighbouring relation?

Label	No (False)
Subject	Accounting and Composition
Topic	Parallel Composition
Negative Mode	atom
Related Question	107
Citation	1997
Section	2.4.2
Publish Year	2009
PDF Link	https://dl.acm.org/doi/10.1145/1810891.1810916 [McSherry, 2010]
Comments	An counter example: Suppose $x_1 < x_2 < \dots < x_n$, then $\mathcal{M}(X) = \{\mathcal{M}_1(x_1), \dots, \mathcal{M}_n(x_n)\}$. Let the neighbouring dataset $X' = \{x_1, x'_2, x_2, \dots, x_{n-1}\}$ and $x_1 < x'_2 < x_2$. Then $\mathcal{M}(X') = \{\mathcal{M}_1(x_1), \mathcal{M}_2(x'_2), \mathcal{M}_3(x_2), \dots, \mathcal{M}_n(x_{n-1})\}$.

By sequential composition, the resulting mechanism satisfies $(\sum_{i=1}^n \varepsilon_i, \sum_{i=1}^n \delta_i)$ -DP.

1.4. Sequential or Adaptive Composition

Question 120

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. For any order $\alpha > 1$, does the composed mechanism $\mathcal{M}$ satisfy $(\alpha, n\varepsilon^2/2)$ -RDP?

Label No (False)

Subject Accounting and Composition

Topic Sequential or Adaptive Composition

Negative Mode conclusion_perturbation

Related Question 122

Citation 2130

Section III, IV

Publish Year 2017

PDF Link <https://arxiv.org/pdf/1702.07476> [Mironov, 2017]

Comments Consider the following counterexample. Let $\mathcal{M}$ be randomized response on a single bit, which outputs the true bit with probability $p = \frac{e^{0.5}}{1+e^{0.5}}$. Then $\mathcal{M}$ is 0.5-DP. However, the statement in question would imply that $\mathcal{M}$ satisfies $(\infty, 0.125)$ -RDP. Since (∞, δ) -RDP is equivalent to δ -DP (because $D_\infty(P\|Q) = \log \sup_y \frac{P(y)}{Q(y)}$), this would further imply that $\mathcal{M}$ is 0.125-DP, contradicting the fact that the privacy loss of randomized response here is exactly 0.5.

Question 121

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(\alpha, \sqrt{n}\varepsilon^2)$ -RDP?

Label No (False)

Subject Accounting and Composition

Topic Sequential or Adaptive Composition

Negative Mode conclusion_perturbation

Related Question 122

Citation 2130

Section III, IV

Publish Year	2017
PDF Link	https://arxiv.org/pdf/1702.07476 [Mironov, 2017]
Comments	Consider the following counterexample. Let $\mathcal{M}$ be randomized response on a single bit, which outputs the true bit with probability $p = \frac{e^{0.5}}{1+e^{0.5}}$. Then $\mathcal{M}$ is 0.5-DP. However, the statement in question would imply that $\mathcal{M}$ satisfies $(\infty, 0.125)$ -RDP. Since (∞, ε) -RDP is equivalent to ε -DP, this would further imply that $\mathcal{M}$ is 0.125-DP, contradicting the fact that the privacy loss of randomized response here is exactly 0.5.

Question 122

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(n\varepsilon^2/2)$ -zCDP

Label	Yes (True)
Subject	Accounting and Composition
Topic	Sequential or Adaptive Composition
Negative Mode	atom
Related Question	122
Citation	1199
Section	1.2.3, 3.1
Publish Year	2016
PDF Link	https://arxiv.org/pdf/1605.02065 [Bun and Steinke, 2016]
Comments	Since each mechanism $\mathcal{M}_i$ satisfies ε -DP, it also satisfies $\frac{\varepsilon^2}{2}$ -zCDP. By the (adaptive) composition property of zCDP, the sequential composition $\mathcal{M}$ therefore satisfies $\frac{n\varepsilon^2}{2}$ -zCDP.

Question 123

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(\sqrt{n}\varepsilon^2/2)$ -zCDP

Label	No (False)
Subject	Accounting and Composition
Topic	Sequential or Adaptive Composition
Negative Mode	conclusion_perturbation
Related Question	122
Citation	1199

Section	1.2.3, 3.1
Publish Year	2016
PDF Link	https://arxiv.org/pdf/1605.02065 [Bun and Steinke, 2016]
Comments	Let the dataset be a single bit $x \in \{0, 1\}$ with neighboring relation $0 \sim 1$. Fix $\varepsilon > 0$, and let each $\mathcal{M}_i$ be randomized response: on input x, output x with probability $p = \frac{e^\varepsilon}{1+e^\varepsilon}$ and output $1-x$ otherwise. Then each $\mathcal{M}_i$ is ε-DP, and the sequential composition $\mathcal{M}$ outputs the vector of n independent randomized-response outputs. Let $P = \mathcal{M}(0)$ and $Q = \mathcal{M}(1)$. For a single draw $i \in [n]$, when $\alpha > 1$, the order-α Rényi divergence is $D_\alpha(P_i \ Q_i) = \frac{1}{\alpha-1} \log\left(\frac{e^{\varepsilon\alpha} + e^{\varepsilon(1-\alpha)}}{1+e^\varepsilon}\right) = \varepsilon - \frac{1}{\alpha-1} \log\left(\frac{1+\exp(-\varepsilon)}{1+\exp((1-2\alpha)\varepsilon)}\right) \geq \varepsilon - \frac{\log(1+\exp(-\varepsilon))}{\alpha-1}$. By additivity under composition, $D_\alpha(P \ Q) = \sum_{i=1}^n D_\alpha(P_i \ Q_i) \geq n\left(\varepsilon - \frac{\log(1+e^{-\varepsilon})}{\alpha-1}\right)$. Choosing $\alpha \geq 1 + \frac{4\log(1+e^\varepsilon)}{\varepsilon}$ gives $D_\alpha(P \ Q) \geq \frac{3}{4}n\varepsilon$. If, towards a contradiction, $\mathcal{M}$ satisfied ρ-zCDP with $\rho = \frac{\sqrt{n}\varepsilon^2}{2}$, then it would satisfy $(\alpha, \alpha\rho)$-RDP for every $\alpha > 1$, hence $D_\alpha(P \ Q) \leq \alpha\rho = \alpha \cdot \frac{\sqrt{n}\varepsilon^2}{2}$. Setting $\alpha = \frac{\sqrt{n}}{\varepsilon}$ (and taking n large enough so that this α also meets the earlier condition) yields the upper bound $\alpha\rho = \frac{n\varepsilon}{2}$, contradicting $D_\alpha(P \ Q) \geq \frac{3}{4}n\varepsilon$. Therefore, in general, the sequential composition of n ε-DP mechanisms does not satisfy $(\frac{\sqrt{n}\varepsilon^2}{2})$-zCDP.

Question 124

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be mechanisms, each satisfying ε/n -DP under the same neighboring relation (assume $\varepsilon < 1$). Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(\frac{\varepsilon}{10}, \exp(\frac{\varepsilon^2}{n} - \frac{\varepsilon}{10} - 1))$ -DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Sequential or Adaptive Composition
Negative Mode	atom
Related Question	124
Citation	1199
Section	2.3
Publish Year	2020
PDF Link	https://arxiv.org/pdf/2004.00010 [Canonne et al., 2020]

Comments	Since each mechanism $\mathcal{M}_i$ satisfies ε/n-DP, it also satisfies $\frac{\varepsilon^2}{2n^2}$-zCDP. By the (adaptive) composition property of zCDP, the sequential composition $\mathcal{M}$ therefore satisfies $\frac{\varepsilon^2}{2n}$-zCDP. Using Canonne et al. [2020, Proposition 12] with $\alpha = 2$ and $\tau = \frac{\varepsilon^2}{n}$, it satisfies $(\frac{\varepsilon}{10}, \delta)$-DP with $\delta = \exp(\frac{\varepsilon^2}{n} - \frac{\varepsilon}{10} - 1)$
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Question 125

Question:

Let $\{\mathcal{M}_i\}_{i=1}^n$ be exponential mechanisms, each satisfying ε/n -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(\frac{\varepsilon}{16}, \exp(\frac{\varepsilon^2}{4n} - \frac{\varepsilon}{16} - 1))$ -DP?

Label	Yes (True)
Subject	Accounting and Composition
Topic	Sequential or Adaptive Composition
Negative Mode	assumption_perturbation
Related Question	124
Citation	1199; 45
Section	2.3; 3
Publish Year	2020; 2020
PDF Link	https://arxiv.org/pdf/2004.00010 https://arxiv.org/abs/2004.07223 [Canonne et al., 2020, Cesar and Rogers, 2020]
Comments	Since each mechanism $\mathcal{M}_i$ satisfies ε/n -DP, by the bounded range property of exponential mechanism, it also satisfies $\frac{\varepsilon^2}{8n^2}$ -zCDP. By the (adaptive) composition property of zCDP, the sequential composition $\mathcal{M}$ therefore satisfies $\frac{\varepsilon^2}{8n}$ -zCDP. Using Canonne et al. [2020, Proposition 12] with $\alpha = 2$ and $\tau = \frac{\varepsilon^2}{n}$, it satisfies $(\frac{\varepsilon}{16}, \delta)$ -DP with $\delta = \exp(\frac{\varepsilon^2}{4n} - \frac{\varepsilon}{16} - 1)$

Question 126**Question:**

Let $\{\mathcal{M}_i\}_{i=1}^n$ be report noisy max mechanisms with gaussian noise, each satisfying $(\varepsilon/n, \delta)$ -DP under the same neighboring relation. Consider their sequential (possibly adaptive) composition on the same dataset. Does the composed mechanism $\mathcal{M}$ satisfy $(\varepsilon, \exp(\frac{\varepsilon^2}{8n} - \varepsilon - 1))$ -DP?

Label	No (False)
Subject	Accounting and Composition
Topic	Sequential or Adaptive Composition
Negative Mode	assumption_perturbation; conclusion_perturbation
Related Question	124
Citation	1199; 45
Section	2.3; 3
Publish Year	2020; 2020
PDF Link	https://arxiv.org/pdf/2004.00010 https://arxiv.org/abs/2004.07223 [Canonne et al., 2020, Cesar and Rogers, 2020]
Comments	Take $n = 1$ and set $\varepsilon = 1$, so $\delta' = \exp(1/8 - 1 - 1) = \exp(-1.875) \approx 0.153$. Choose $\delta = 0.2 > \delta'$, and let $\mathcal{M}$ be a Gaussian report-noisy-max mechanism

that satisfies $(1, 0.2)$ -DP. Since (ε, δ) -DP does not imply (ε, δ') -DP for any $\delta' < \delta$, it follows that $\mathcal{M}$ need not satisfy $(1, 0.153)$ -DP.

1.5. SVT

Question 130

Question:

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP? (assume $c > 1$)

Algorithm 1: SVT

- 1 **Input.** A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^\infty$ each with sensitivity at most Δ , a set of thresholds $\{T_i\}_{i=1}^\infty$, privacy budget ε , and an answer cap $c \in \mathbb{N}$.
 - 2 **Procedure.**
 1. Split privacy budget: $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$.
 2. Sample a noisy reference threshold $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. Initialize counter $r \leftarrow 0$.
 4. For each query q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(2c\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T_i + \theta$, output $\top$ and increment r .
 - (c) Stop if $r \geq c$; otherwise continue.
 - (d) If the condition fails, output $\perp$.
-

Label	Yes (True)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	None
Related Question	None
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 1]

Question 131

Question:

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP? (assume $c > 1$)

Algorithm 2: SVT

-
- 1 **Input.** A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ each with sensitivity at most Δ , a thresholds T , privacy budget ε , and an answer cap $c \in \mathbb{N}$.
 - 2 **Procedure.**
 1. Allocate $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$.
 2. Sample threshold noise $\theta \sim \text{Lap}(c\Delta/\varepsilon_1)$.
 3. Initialize $r \leftarrow 0$.
 4. For each q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(2c\Delta/\varepsilon_1)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T + \theta$, report $\top$.
 - (c) Resample $\theta' \sim \text{Lap}(c\Delta/\varepsilon_2)$ for the next iteration.
 - (d) Increment r , and stop once $r \geq c$.
 - (e) Otherwise output $\perp$.
-

Label	Yes (True)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	None
Related Question	None
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 2]

Question 132**Question:**

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP? (assume $c > 1$)

Algorithm 3: SVT

-
- 1 **Input.** A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ each with sensitivity at most Δ , a thresholds T , privacy budget ε , and an answer cap $c \in \mathbb{N}$.
 - 2 **Procedure.**
 1. Split ε as $\varepsilon_1 = \varepsilon/2$, $\varepsilon_2 = \varepsilon - \varepsilon_1$.
 2. Sample threshold offset $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. Initialize $r \leftarrow 0$.
 4. For each q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(c\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T + \theta$, release the noisy score $q_i(\mathcal{D}) + \eta_i$.
 - (c) Increment r and stop if $r \geq c$.
 - (d) Otherwise, return $\perp$.
-

Label	No (False)
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Subject	Accounting and Composition
Topic	SVT
Negative Mode	natural_error
Related Question	None
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 3]

Question 133

Question:

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP? (assume $c > 1$)

Algorithm 4: SVT

- 1 **Input.** A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ each with sensitivity at most Δ , a thresholds T , privacy budget ε , and an answer cap $c \in \mathbb{N}$.
- 2 **Procedure.**
 1. Allocate $\varepsilon_1 = \varepsilon/4$ to threshold noise; $\varepsilon_2 = \varepsilon - \varepsilon_1$.
 2. Draw $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$ and set $r \leftarrow 0$.
 3. For each q_i :
 - (a) Sample $\eta_i \sim \text{Lap}(\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T + \theta$, emit $\top$ and increment r .
 - (c) Otherwise output $\perp$; stop when $r \geq c$;

Label	No (False)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	natural_error
Related Question	None
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 4]

Question 134

Question:

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP?

Algorithm 5: SVT

- 1 **Input.** Database $\mathcal{D}$, queries Q , sensitivity Δ , global threshold T , privacy budget ε .
 - 2 **Procedure.**
 1. Use $\varepsilon_1 = \varepsilon/2$, $\varepsilon_2 = \varepsilon - \varepsilon_1$.
 2. Perturb threshold once: $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. Set $r \leftarrow 0$.
 4. For each q_i :
 - (a) Output $\top$ if $q_i(\mathcal{D}) \geq T + \theta$, otherwise $\perp$.
-

Label	No (False)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	natural_error
Related Question	None
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 5]

Question 135
Question:

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP?

Algorithm 6: SVT

- 1 **Input.** Dataset $\mathcal{D}$, sensitivity Δ , per-query thresholds $\{T_i\}$, and total privacy budget ε .
 - 2 **Procedure.**
 1. Divide budget: $\varepsilon_1 = \varepsilon/2$, $\varepsilon_2 = \varepsilon - \varepsilon_1$.
 2. Sample $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. For each q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T_i + \theta$, output $\top$; else $\perp$.
-

Label	No (False)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	natural_error
Related Question	None
Citation	212
Section	3

Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Lyu et al. [2017, Algorithm 6]

Question 136**Question:**

Does the following implementation of the sparse vector technique (SVT) satisfy $\varepsilon/2$ -DP? (assume $c > 1$)

Algorithm 7: SVT

-
- 1 **Input.** A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ each with sensitivity at most Δ , a set of thresholds $\{T_i\}_{i=1}^{\infty}$, privacy budget ε , and an answer cap $c \in \mathbb{N}$.
 - 2 **Procedure.**
 1. Split privacy budget: $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$.
 2. Sample a noisy reference threshold $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. Initialize counter $r \leftarrow 0$.
 4. For each query q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(2c\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq T_i + \theta$, output $\top$ and increment r .
 - (c) Stop if $r \geq c$; otherwise continue.
 - (d) If the condition fails, output $\perp$.
-

Label	No (False)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	conclusion_perturbation
Related Question	130
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	The privacy guarantee comes from Lyu et al. [2017, Algorithm 1], should be ε -DP by adaptive composition.

Question 137**Question:**

Does the following implementation of the sparse vector technique (SVT) satisfy ε -DP, under the add/remove neighbouring relation? (assume $c > 1$)

Algorithm 8: SVT

-
- 1 Input.** A database $\mathcal{D}$ with each datum $x \in \mathbb{R}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ with $q_i = \sum_{x \in \mathcal{D}} \mathbf{1}\{x \geq i\}$, privacy budget ε , and an answer cap $c \in \mathbb{N}$.
- 2 Procedure.**
1. Split privacy budget: $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$.
 2. Sample a noisy reference threshold $\theta \sim \text{Lap}(\Delta/\varepsilon_1)$.
 3. Initialize counter $r \leftarrow 0$.
 4. For each query q_i :
 - (a) Draw $\eta_i \sim \text{Lap}(c\Delta/\varepsilon_2)$.
 - (b) If $q_i(\mathcal{D}) + \eta_i \geq i + \theta$, output $\top$ and increment r .
 - (c) Stop if $r \geq c$; otherwise continue.
 - (d) If the condition fails, output $\perp$.
-

Label	Yes (True)
Subject	Accounting and Composition
Topic	SVT
Negative Mode	assumption_perturbation
Related Question	130
Citation	212
Section	3
Publish Year	2016
PDF Link	https://arxiv.org/abs/1603.01699 [Lyu et al., 2017]
Comments	Algorithm 1 in Lyu et al. [2017]. Moreover, by the monotone query property, the noise in line 4(a) only needs to be $\text{Lap}(c\Delta/\varepsilon_2)$ (rather than $\text{Lap}(2c\Delta/\varepsilon_2)$).

2. DP-statistics**2.1. DP selection: expoMech****Question 18****Question:**

The pseudocode below implements the exponential mechanism. Does this implementation ensure ε -differential privacy under the add/remove neighboring relationship?

Algorithm 9: Exponential Mechanism

-
- Input :** Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y ; utility (score) function $q(D, y) = \sum_{i=1}^n \mathbf{1}\{z_i = y\}$; privacy parameter ε
- Output:** Selected item $\hat{y} \in Y$ sampled by the exponential mechanism
-
- 1 for** $y \in Y$ **do**
 - 2** $p_y \leftarrow \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}$
 - 3** Sample $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$
 - 4 return** $\hat{y}$
-

Label	Yes (True)
Subject	DP-statistics
Topic	DP selection: expoMech
Negative Mode	atom
Related Question	18
Citation	80
Section	3
Publish Year	2019
PDF Link	https://arxiv.org/pdf/1909.13830 [Dong et al., 2019]
Comments	For a monotone query q , the exponential mechanism with $p_y \propto \exp\left(\frac{\varepsilon q}{\Delta_q}\right)$ satisfies ε -DP [Dong et al., 2019].

Question 19

Question:

The pseudocode below implements the exponential mechanism. Does this implementation ensure ε -differential privacy under the add/remove neighboring relationship?

Algorithm 10: Exponential Mechanism

Input : Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y ; utility (score) function

$q(D, y) = \left| \sin\left(\frac{\pi}{2} \sum_{i=1}^n \mathbf{1}\{z_i = y\}\right) \right|$; privacy parameter ε

Output: Selected item $\hat{y} \in Y$ sampled by the exponential mechanism

```

1 for  $y \in Y$  do
2    $p_y \leftarrow \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}$ 
3 Sample  $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$ 
4 return  $\hat{y}$ 

```

Label	Yes (True)
Subject	DP-statistics
Topic	DP selection: expoMech
Negative Mode	assumption_perturbation
Related Question	18
Citation	80
Section	3
Publish Year	2019
PDF Link	https://arxiv.org/pdf/1909.13830 [Dong et al., 2019]
Comments	Let Y be finite candidate set. Given dataset $D = \{z_i\}_{i=1}^n$ denote $c_y(D) := \sum_{i=1}^n \mathbf{1}\{z_i = y\}$, then $q(D, y) = \left \sin\left(\frac{\pi}{2} c_y(D)\right) \right $. Consider mechanism M sampling $y \in Y$ with $p_D(y) := \Pr[M(D) = y] = \frac{\exp(\varepsilon q(D, y))}{Z(D)}$, where $Z(D) := \sum_{u \in Y} \exp(\varepsilon q(D, u))$. Notice that $\sin(\frac{\pi}{2} c_y(D)) \in \{0, \pm 1\}$, hence

$q(D, y) \in \{0, 1\}$. Without loss of generality, let $D' = D \cup \{z^*\}$ or $D = D' \cup \{z^*\}$ and set $y^* := z^*$. Then c_{y^*} changes by ± 1 and hence $q(\cdot, y^*)$ flips between 0 and 1, while for all $y \neq y^*$ we have $q(D, y) = q(D', y)$. Therefore for all $y \neq y^*$, $\frac{p_D(y)}{p_{D'}(y)} = \frac{Z(D')}{Z(D)}$. Moreover, since $q \in \{0, 1\}$, $Z(D) = |Y| + (e^\varepsilon - 1)|A(D)|$ where $A(D) := \{y : q(D, y) = 1\}$, and $|A(D')| - |A(D)| \in \{-1, +1\}$, implying $e^{-\varepsilon} \leq \frac{Z(D')}{Z(D)} \leq e^\varepsilon$. Hence $\frac{p_D(y)}{p_{D'}(y)} \leq e^\varepsilon$ for all $y \neq y^*$. For $y = y^*$, the numerator changes by at most a factor e^ε , i.e., $\exp(\varepsilon q(D, y^*)) / \exp(\varepsilon q(D', y^*)) \leq e^\varepsilon$, and thus $\frac{p_D(y^*)}{p_{D'}(y^*)} = \frac{\exp(\varepsilon q(D, y^*))}{\exp(\varepsilon q(D', y^*))} \cdot \frac{Z(D')}{Z(D)} \leq e^\varepsilon$. Therefore $p_D(y) \leq e^\varepsilon p_{D'}(y)$ for all $y \in Y$. The reverse direction is symmetric, so M is ε -DP.

Question 20

Question:

The pseudocode below implements the exponential mechanism. Does this implementation ensure ε -differential privacy under the replace-one neighboring relationship?

Algorithm 11: Exponential Mechanism

Input : Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y ; utility (score) function

$$q(D, y) = \sum_{i=1}^n \mathbf{1}\{z_i = y\}; \text{ privacy parameter } \varepsilon$$

Output: Selected item $\hat{y} \in Y$ sampled by the exponential mechanism

```

1 for  $y \in Y$  do
2    $p_y \leftarrow \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}$ 
3 Sample  $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$ 
4 return  $\hat{y}$ 
```

Label	No (False)
Subject	DP-statistics
Topic	DP selection: expoMech
Negative Mode	conclusion_perturbation
Related Question	18
Citation	83
Section	3
Publish Year	2024
PDF Link	https://arxiv.org/pdf/1909.13830 [Dong et al., 2019]
Comments	The exponent in the density of the exponential mechanism is missing a factor of $1/2$. Consequently, under the replace-one neighboring relation, it satisfies only 2ε -DP.

Question 21

Question:

The pseudocode below implements the exponential mechanism. Does this implementation ensure $\varepsilon/2$ -differential privacy under the add/remove neighboring relationship?

Algorithm 12: Exponential Mechanism

Input : Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y ; utility (score) function $q(D, y) = \sum_{i=1}^n \mathbf{1}\{z_i = y\}$; privacy parameter ε
Output: Selected item $\hat{y} \in Y$ sampled by the exponential mechanism

```

1 for  $y \in Y$  do
2    $p_y \leftarrow \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}$ 
3 Sample  $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$ 
4 return  $\hat{y}$ 

```

Label No (False)

Subject DP-statistics

Topic DP selection: expoMech

Negative Mode conclusion_perturbation

Related Question 18

Citation 80

Section 3

Publish Year 2019

PDF Link <https://arxiv.org/pdf/1909.13830> [Dong et al., 2019]

Comments For a monotone query q , the exponential mechanism with $p_y \propto \exp\left(\frac{\varepsilon q}{\Delta_q}\right)$ is ε -DP [Dong et al., 2019], which is weaker than the guarantee requested in the statement, namely $\varepsilon/2$ -DP.

Question 22

Question:

The pseudocode below implements the exponential mechanism. Does this implementation ensure ε -differential privacy under the add/remove neighboring relationship?

Algorithm 13: Exponential Mechanism

Input : Dataset $D = \{z_i\}_{i=1}^n$; candidate set Y ; utility (score) function $q(D, y) = \sum_{i=1}^n \mathbf{1}\{z_i = y\}$; privacy parameter ε
Output: Selected item $\hat{y} \in Y$ sampled by the exponential mechanism

```

1 for  $y \in Y$  do
2    $p_y \leftarrow \frac{\exp(2\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(2\varepsilon q(D, y'))}$ 
3 Sample  $\hat{y} \sim \text{Multinomial}(1; \{p_y\}_{y \in Y})$ 
4 return  $\hat{y}$ 

```

Label No (False)

Subject DP-statistics

Topic DP selection: expoMech

Negative Mode	algorithm_perturbation
Related Question	18
Citation	80
Section	3
Publish Year	2019
PDF Link	https://arxiv.org/pdf/1909.13830 [Dong et al., 2019]
Comments	For a monotone query q , the exponential mechanism with $p_y \propto \exp\left(\frac{\varepsilon q}{\Delta_q}\right)$ is ε -DP [Dong et al., 2019], thus the algorithm described above satisfies 2ε -DP.

2.2. mean estimation

Question 72

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 14: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping thresld C

- 1 Sample $\xi_1 \sim \text{Lap}(2C/\varepsilon)$
 - 2 Sample $\xi_2 \sim \text{Lap}(2/\varepsilon)$
 - 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
 - 4 **return** $\hat{\theta}$
-

Label	Yes (True)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	atom
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Under the add/remove neighboring relation, the numerator has sensitivity C and the denominator has sensitivity 1. By (sequential) composition, the resulting released estimator is ε -DP.

Question 73

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 15: Mean estimation**Input** : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi \sim \text{Lap}(C/\varepsilon)$
- 2 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\}}{n} + \xi$
- 3 **return** $\hat{\theta}$

Label Yes (True)**Subject** DP-statistics**Topic** mean estimation**Negative Mode** algorithm_perturbation**Related Question** 72**Citation** None**Section** None**Publish Year** None

Comments Assuming $0/0 = 0$, under the add/remove neighboring relation, the sensitivity of $\frac{\sum_{i=1}^n x_i}{n}$ is upper bounded by $\max\left\{\frac{2C}{n+1}, \frac{2C}{n-1}\right\}$. The first term is at most C , while the second term is maximized at $n = 2$. When $n = 2$, however, the sensitivity under the remove-one relation admits the tighter bound C . Therefore, it suffices to add Laplace noise calibrated to C .

Question 74**Question:**

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship? (assume $2 < n < 10$)

Algorithm 16: Mean estimation**Input** : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi \sim \text{Lap}(C/\varepsilon)$
- 2 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + 10\xi}{n}$
- 3 **return** $\hat{\theta}$

Label Yes (True)**Subject** DP-statistics**Topic** mean estimation**Negative Mode** assumption_perturbation; algorithm_perturbation**Related Question** 72**Citation** None**Section** None**Publish Year** None

Comments The sensitivity is upper bounded by $\max\left\{\frac{2C}{n+1}, \frac{2C}{n-1}\right\}$. Hence it suffices to check the cases $n = 3$ and $n = 9$, which clearly satisfy ε -DP since the mechanism adds more noise than required.

Question 75

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 17: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}(2C/\varepsilon)$
- 2 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n}$
- 3 **return** $\hat{\theta}$

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	assumption_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Under the add/remove neighboring relation, the sample size n is private, and the proposed estimator does not add enough noise to privatize n .

Question 76

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 18: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}(2C/\varepsilon)$
- 2 Sample $\xi_2 \sim \text{Lap}(2/\varepsilon)$
- 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^{n-1} \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
- 4 **return** $\hat{\theta}$

Label	No (False)
Subject	DP-statistics
Topic	mean estimation

Negative Mode	algorithm_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Because the sum ranges from 1 to $n - 1$, it can induce sensitivity $2C$ under the add/remove neighboring relation. For instance, consider the neighboring datasets $\{C, -C, C\}$ and $\{C, -C, -C, C\}$.

Question 77

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 19: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}(2C/\varepsilon)$
 - 2 Sample $\xi_2 \sim \text{Lap}(1/\varepsilon)$
 - 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
 - 4 **return** $\hat{\theta}$
-

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	algorithm_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Under the add/remove neighboring relation, the numerator has sensitivity C and the denominator has sensitivity 1. By (sequential) composition, the resulting released estimator is $\frac{3\varepsilon}{2}$ -DP.

Question 78

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 20: Mean estimation**Input** : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.**Output:** Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{(x - l)/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 $\tilde{m} \leftarrow l + \Delta \cdot \rho$.
- 8 **return** $\tilde{m}$.

Label	Yes (True)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	atom
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]
Comments	The algorithm is given in Kulesza et al. [2023, Algorithm 3], and its privacy guarantee is proved in Kulesza et al. [2023, Theorem 1].

Question 79**Question:**

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 21: Mean estimation**Input** : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.**Output:** Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{x/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 $\tilde{m} \leftarrow l + \Delta \cdot \rho$.
- 8 **return** $\tilde{m}$.

Label	No (False)
--------------	------------

Subject	DP-statistics
Topic	mean estimation
Negative Mode	algorithm_perturbation
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]
Comments	The algorithm is modified from Kulesza et al. [2023, Algorithm 3]. However, it is not ε -DP because $ a(X) - a(X') = x/\Delta $ need not be bounded by 1, so the noise scale of η_1 may be insufficient.

Question 80

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 22: Mean estimation

Input : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.

Output: Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{(x - u)/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 return $\tilde{m}$.

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	algorithm_perturbation
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]

Comments The algorithm is modified from Kulesza et al. [2023, Algorithm 3]. However, it is not ε -DP since $|b(X) - b(X')| = \left|1 + \frac{u-x}{u-l}\right| \geq 1$, so the noise scale of η_1 may be insufficient.

Question 81

Question:

Does the following algorithm satisfy $\varepsilon/2$ -differential privacy under the add/remove neighboring relationship?

Algorithm 23: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}(2C/\varepsilon)$
- 2 Sample $\xi_2 \sim \text{Lap}(2/\varepsilon)$
- 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
- 4 **return** $\hat{\theta}$

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	conclusion_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Under the add/remove neighboring relation, the numerator has sensitivity C and the denominator has sensitivity 1. By (sequential) composition, the resulting released estimator is ε -DP.

Question 82

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 24: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}((C+1)/\varepsilon)$
- 2 Sample $\xi_2 \sim \text{Lap}((C+1)/\varepsilon)$
- 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
- 4 **return** $\hat{\theta}$

Label	Yes (True)
Subject	DP-statistics

Topic	mean estimation
Negative Mode	algorithm_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	Under the add/remove neighboring relation, the L_1 sensitivity of releasing the vector $\left(\sum_{i=1}^n \text{Clip}(x_i), n\right)$ is $C + 1$. Hence, adding two-dimensional Laplace noise with per-coordinate scale $(C + 1)/\varepsilon$ achieves ε -DP.

Question 83

Question:

Does the following algorithm satisfy ε -differential privacy under the add/remove neighboring relationship?

Algorithm 25: Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\}$, privacy budget ε , clipping threshold C

- 1 Sample $\xi_1 \sim \text{Lap}((C + 1)/\varepsilon)$
 - 2 Let $\xi_2 \leftarrow \xi_1$
 - 3 Let $\hat{\theta} \leftarrow \frac{\sum_{i=1}^n \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$
 - 4 **return** $\hat{\theta}$
-

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	algorithm_perturbation
Related Question	72
Citation	None
Section	None
Publish Year	None
Comments	The added Laplace noises ξ_1 and ξ_2 are identical and not independent, thus not guarantee ε -DP.

Question 84

Question:

Does the following algorithm satisfy ε -differential privacy under the replace-one neighboring relationship?

Algorithm 26: Mean estimation**Input** : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.**Output:** Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{(x - l)/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 $\tilde{m} \leftarrow l + \Delta \cdot \rho$.
- 8 **return** $\tilde{m}$.

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	assumption_perturbation
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]
Comments	The algorithm appears as Kulesza et al. [2023, Algorithm 3], and its privacy guarantee is proved in Kulesza et al. [2023, Theorem 1] under the add/remove neighboring relation. Since the statement here asks for a guarantee under the replace-one neighboring relation, the noise scale should be doubled.

Question 85**Question:**

Does the following algorithm satisfy $(\alpha, 0.8\varepsilon)$ -RDP under the add/remove neighboring relationship, for any $\alpha > 1$?

Algorithm 27: Mean estimation**Input** : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.**Output:** Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{(x - l)/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 $\tilde{m} \leftarrow l + \Delta \cdot \rho$.
- 8 **return** $\tilde{m}$.

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	conclusion_perturbation
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]
Comments	Taking $\alpha = \infty$, the statement reduces to checking whether the algorithm satisfies 0.8ε -DP. Under the add/remove neighboring relation, Kulesza et al. [2023, Theorem 1] proves that the algorithm is ε -DP. Thus, the statement is not true.

Question 86**Question:**

Does the following algorithm satisfy $(\alpha, \frac{(3\alpha^2+11\alpha+8)\varepsilon}{5\alpha^2+6\alpha})$ -RDP under the add/remove neighboring relationship, for any $\alpha > 1$?

Algorithm 28: Mean estimation**Input :** Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.**Output:** Noisy mean estimate $\tilde{m}$.

- 1 Set $\Delta \leftarrow u - l$.
- 2 Let $\mathcal{X} \leftarrow \{(x - l)/\Delta : x \in \mathcal{D}\}$.
- 3 $a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$.
- 4 Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, ξ_1 and ξ_2 are i.i.d. $\text{Lap}(1/\varepsilon)$
- 5 $\hat{a} \leftarrow a + \eta_1$, $\hat{b} \leftarrow b + \eta_2$.
- 6 $\rho \leftarrow \min\{\max\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\}, 1\}$.
- 7 $\tilde{m} \leftarrow l + \Delta \cdot \rho$.
- 8 **return** $\tilde{m}$.

Label	No (False)
Subject	DP-statistics
Topic	mean estimation
Negative Mode	conclusion_perturbation
Related Question	78
Citation	8
Section	3
Publish Year	2023
PDF Link	https://arxiv.org/abs/2312.06658 [Kulesza et al., 2023]
Comments	Taking $\alpha = \infty$, the statement reduces to checking whether the algorithm satisfies 0.8ε -DP. Under the add/remove neighboring relation, Kulesza et al. [2023, Theorem 1] proves that the algorithm is ε -DP. Thus, the statement is not true.

2.3. quantile**Question 115****Question:**Does the quantile estimation mechanism described below satisfy ε -differential privacy?**Algorithm 29:** DP Single Quantile Estimation via Exponential Mechanism**Input:** Quantile level $q \in [0, 1]$, dataset $X = (x_1, \dots, x_n) \subset (a, b)^n$, privacy budget ε .**Output:** Quantile estimate $v \in (a, b)$.

- 1 Define: $u(X, w) = -|\#\{x \in X \mid x < w\} - \lfloor qn \rfloor|$
- 2 Partition (a, b) into intervals $I_k = [x_{k-1}, x_k]$, $k = 1, \dots, n+1$ where $x_0 = a$, $x_{n+1} = b$
- 3 Sample I_k with probability

$$\Pr[\mathcal{A}(X) = I_k] \propto \exp\left(\frac{\varepsilon u(X, I_k)}{2}\right) (x_k - x_{k-1}).$$

- 4 Output $v \sim \text{Unif}(I_k)$.

Label	Yes (True)
Subject	DP-statistics
Topic	quantile
Negative Mode	atom
Related Question	115
Citation	32
Section	appendix A
Publish Year	2021
PDF Link	https://proceedings.mlr.press/v162/kaplan22a/kaplan22a.pdf [Kaplan et al., 2022]
Comments	The algorithm is the DP single-quantile mechanism defined in Kaplan et al. [2022, Appendix A]. It satisfies ε -DP by Kaplan et al. [2022, Lemma A.1].

Question 116

Question:

Does the quantile estimation mechanism described below satisfy $\varepsilon/2$ -differential privacy?

Algorithm 30: DP Single Quantile Estimation via Exponential Mechanism

Input: Quantile level $q \in [0, 1]$, dataset $X = (x_1, \dots, x_n) \subset (a, b)^n$, privacy budget ε .

Output: Quantile estimate $v \in (a, b)$.

- 1 Define: $u(X, w) = -|\#\{x \in X \mid x < w\} - \lfloor qn \rfloor|$
- 2 Partition (a, b) into intervals $I_k = [x_{k-1}, x_k]$, $k = 1, \dots, n+1$ where $x_0 = a$, $x_{n+1} = b$
- 3 Sample I_k with probability

$$\Pr[\mathcal{A}(X) = I_k] \propto \exp\left(\frac{\varepsilon u(X, I_k)}{2}\right)(x_k - x_{k-1}).$$

- 4 Output $v \sim \text{Unif}(I_k)$.
-

Label	No (False)
Subject	DP-statistics
Topic	quantile
Negative Mode	conclusion_perturbation
Related Question	115
Citation	32
Section	appendix A
Publish Year	2021
PDF Link	https://proceedings.mlr.press/v162/kaplan22a/kaplan22a.pdf [Kaplan et al., 2022]
Comments	The algorithm is the DP single-quantile mechanism defined in Kaplan et al. [2022, Appendix A]. It satisfies ε -DP by Kaplan et al. [2022, Lemma A.1].

Question 117**Question:**

Does the quantile estimation mechanism described below satisfy ε -differential privacy under add/remove neighbouring relation?

Algorithm 31: DP Single Quantile Estimation via Exponential Mechanism

Input: Quantile level $q \in [0, 1]$, dataset $X = (x_1, \dots, x_n) \subset (a, b)^n$, privacy budget ε .

Output: Quantile estimate $v \in (a, b)$.

- 1 Define: $u(X, w) = -|\#\{x \in X \mid x < w\} - \lfloor qn \rfloor|$
- 2 Partition (a, b) into intervals $I_k = [x_{k-1}, x_k]$, $k = 1, \dots, n+1$ where $x_0 = a$, $x_{n+1} = b$
- 3 Sample I_k with probability

$$\Pr[\mathcal{A}(X) = I_k] \propto \exp\left(\frac{\varepsilon u(X, I_k)}{2}\right) (x_k - x_{k-1}).$$

- 4 Output $v \sim \text{Unif}(I_k)$.
-

Label	Yes (True)
Subject	DP-statistics
Topic	quantile
Negative Mode	assumption_perturbation
Related Question	115
Citation	32
Section	appendix A
Publish Year	2021
PDF Link	https://proceedings.mlr.press/v162/kaplan22a/kaplan22a.pdf [Kaplan et al., 2022]
Comments	The algorithm is the DP single-quantile mechanism defined in Kaplan et al. [2022, Appendix A]. It satisfies ε -DP by Kaplan et al. [2022, Lemma A.1].

Question 118**Question:**

Does the quantile-estimation mechanism described below satisfy ε -differential privacy?

Algorithm 32: DP Single Quantile Estimation via Exponential Mechanism**Input:** Quantile level $q \in [0, 1]$, dataset $X = (x_1, \dots, x_n) \subset (a, b)^n$, privacy budget ϵ .**Output:** Quantile estimate $v \in (a, b)$.

- 1 Define: $u(X, w) = -|\#\{x \in X \mid x < w\} - \lfloor qn \rfloor|$
- 2 Let $I_k = [x_{k-1}, x_k]$, $k = 1, \dots, n+1$ where $x_0 = a$, $x_{n+1} = b$
- 3 Sample I_k with probability

$$\Pr[\mathcal{A}(X) = I_k] \propto \exp\left(\epsilon u(X, I_k)\right)(x_k - x_{k-1}).$$

- 4 Output $v \sim \text{Unif}(I_k)$.

Label	No (False)
Subject	DP-statistics
Topic	quantile
Negative Mode	algorithm_perturbation
Related Question	83.0
Citation	32
Section	appendix A
Publish Year	2021
PDF Link	https://proceedings.mlr.press/v162/kaplan22a/kaplan22a.pdf [Kaplan et al., 2022]
Comments	The algorithm is modified from the DP single-quantile mechanism defined in Kaplan et al. [2022, Appendix A]. However, the exponential mechanism is missing a factor of 1/2 in the exponent of its density and therefore does not satisfy ϵ -DP.

Question 119**Question:**Does the quantile-estimation mechanism described below satisfy ϵ -differential privacy?**Algorithm 33:** DP Single Quantile Estimation via Exponential Mechanism**Input:** Quantile level $q \in [0, 1]$, dataset $X = (x_1, \dots, x_n) \subset (a, b)^n$, privacy budget ϵ .**Output:** Quantile estimate $v \in (a, b)$.

- 1 Define: $u(X, w) = -|\#\{x \in X \mid x < w\} - \lfloor qn \rfloor|$
- 2 Let $I_k = [x_{k-1}, x_k]$, $k = 1, \dots, n+1$ where $x_0 = a$, $x_{n+1} = b$
- 3 Sample I_k using exponential mechanism

$$\Pr[\mathcal{A}(X) = I_k] \propto \exp\left(\frac{\epsilon u(X, I_k)}{2}\right).$$

- 4 Output $v \sim \text{Unif}(I_k)$.

Label	No (False)
Subject	DP-statistics
Topic	quantile

Negative Mode	algorithm_perturbation
Related Question	84.0
Citation	32
Section	appendix A
Publish Year	2021
PDF Link	https://proceedings.mlr.press/v162/kaplan22a/kaplan22a.pdf [Kaplan et al., 2022]
Comments	The algorithm is modified from the DP single-quantile mechanism defined in Kaplan et al. [2022, Appendix A]. However, the exponential mechanism is missing the interval-length multiplier in its density and therefore does not satisfy ε -DP.

3. DP-ML

3.1. dp-adam

Question 23

Question:

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson subsampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Could the following implementation of DP-Adam with appropriate noise multiplier σ satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$ (under add/remove neighbouring relation)?

Algorithm 34: DP-Adam

Input: Data $\mathcal{D} = \{x_i\}_{i=1}^N$; epochs E ; batch size B ; learning rate η ; clipping threshold C ;
 initial parameters $\theta_0 \in \mathbb{R}^p$; initial moments $m_0, v_0 \in \mathbb{R}^p$; decay rates $\beta_1, \beta_2 \in (0, 1)$;
 small constant $\gamma > 0$.

- 1 **Set** $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{B}{N}, \lfloor E \cdot N/B \rfloor)$
- 2 **for** $t = 0, 1, \dots, \lfloor E \cdot N/B \rfloor - 1$ **do**
- 3 **Sample** a minibatch B_t via Poisson sampling, where each element of $\mathcal{D}$ is included
 independently with probability B/N .
- 4 **for** $x_i \in B_t$ **do**
- 5 $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(x_i)$
- 6 $\tilde{g}_t(x_i) \leftarrow g_t(x_i) \cdot \min(1, \frac{C}{\|g_t(x_i)\|_2})$
- 7 $z_t \sim \mathcal{N}(0, \sigma^2 C^2 I_p)$
- 8 $\bar{g}_t \leftarrow \frac{1}{B} (\sum_{x_i \in B_t} \tilde{g}_t(x_i) + z_t)$
- 9 $m_{t+1} \leftarrow \beta_1 m_t + (1 - \beta_1) \bar{g}_t$
- 10 $v_{t+1} \leftarrow \beta_2 v_t + (1 - \beta_2) \bar{g}_t^{\odot 2}$ // $\odot$ is elementwise square
- 11 $\hat{m}_{t+1} \leftarrow \frac{m_{t+1}}{1 - \beta_1^{t+1}}$
- 12 $\hat{v}_{t+1} \leftarrow \frac{v_{t+1}}{1 - \beta_2^{t+1}}$
- 13 $\theta_{t+1} \leftarrow \theta_t - \eta \frac{\hat{m}_{t+1}}{\sqrt{\hat{v}_{t+1} + \gamma}}$
- 14 **return** $\theta_{\lfloor E \cdot N/B \rfloor}$

Label	Yes (True)
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Subject	DP-ML
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Topic	dp-adam
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Negative Mode	None
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Related Question	100.0
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Citation	509
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Section	appendix A
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Publish Year	2021
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PDF Link	https://arxiv.org/abs/2110.05679 [Li et al., 2021]
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Comments	The algorithm is DP-Adam, adapted from Li et al. [2021, Algos. 1–2]. Overall, it is essentially a Poisson-subsampled Gaussian mechanism followed by post-processing. Therefore, once the noise multiplier is calibrated using Poisson-subsampled DP-SGD, the entire algorithm satisfies (ε, δ) -DP.
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Question 24**Question:**

Suppose we have a privacy accountant $\mathcal{O}$. Given target privacy parameters (ε, δ) , a Poisson subsampling rate q , and T DP-SGD iterations (i.e., T compositions), $\mathcal{O}$ returns the smallest noise multiplier σ for the subsampled Gaussian mechanism in DP-SGD (under the add/remove neighboring relation) that guarantees (ε, δ) -DP. Now given privacy budget $\varepsilon > 0$ and $\delta \in (0, 1)$,

does the following implementation of DP-Adam satisfy (ε, δ) -DP (under replace-one neighbouring relation)?

Algorithm 35: DP-Adam

Input: Data $\mathcal{D} = \{x_i\}_{i=1}^N$; epochs E ; batch size B ; learning rate η ; clipping threshold C ; initial parameters $\theta_0 \in \mathbb{R}^p$; initial moments $m_0, v_0 \in \mathbb{R}^p$; decay rates $\beta_1, \beta_2 \in (0, 1)$; small constant $\gamma > 0$.

```

1 Set  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{B}{N}, \lfloor E \cdot N/B \rfloor)$ 
2 for  $t = 0, 1, \dots, \lfloor E \cdot N/B \rfloor - 1$  do
3   Sample a minibatch  $B_t$  via Poisson sampling, where each element of  $\mathcal{D}$  is included
     independently with probability  $B/N$ .
4   for  $x_i \in B_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(x_i)$ 
6      $\tilde{g}_t(x_i) \leftarrow g_t(x_i) \cdot \min(1, \frac{C}{\|g_t(x_i)\|_2})$ 
7    $z_t \sim \mathcal{N}(0, \sigma^2 C^2 I_p)$ 
8    $\bar{g}_t \leftarrow \frac{1}{B} (\sum_{x_i \in B_t} \tilde{g}_t(x_i) + z_t)$ 
9    $m_{t+1} \leftarrow \beta_1 m_t + (1 - \beta_1) \bar{g}_t$ 
10   $v_{t+1} \leftarrow \beta_2 v_t + (1 - \beta_2) \bar{g}_t^{\odot 2}$  //  $\odot$  is elementwise square
11   $\hat{m}_{t+1} \leftarrow \frac{m_{t+1}}{1 - \beta_1^{t+1}}$ 
12   $\hat{v}_{t+1} \leftarrow \frac{v_{t+1}}{1 - \beta_2^{t+1}}$ 
13   $\theta_{t+1} \leftarrow \theta_t - \eta \frac{\hat{m}_{t+1}}{\sqrt{\hat{v}_{t+1} + \gamma}}$ 
14 return  $\theta_{\lfloor E \cdot N/B \rfloor}$ 

```

Label No (False)

Subject DP-ML

Topic dp-adam

Negative Mode assumption_perturbation

Related Question 100.0

Citation 509

Section appendix A

Publish Year 2021

PDF Link <https://arxiv.org/abs/2110.05679> [Li et al., 2021]

Comments Since the hypothetical oracle $\mathcal{O}$ returns the smallest noise multiplier calibrated for the add/remove neighboring relation, using the same multiplier under the replace-one neighboring relation generally underestimates the required noise scale. Thus the algorithm doesn't satisfy stated privacy guarantee.

Question 25

Question:

Suppose we have a privacy accountant $\mathcal{O}$. Given target privacy parameters (ε, δ) , a Poisson subsampling rate q , and T DP-SGD iterations (i.e., T compositions), $\mathcal{O}$ returns the smallest noise multiplier σ for the subsampled Gaussian mechanism in DP-SGD (under the add/remove neighboring relation) that guarantees (ε, δ) -DP. Now given privacy budget $\varepsilon > 0$ and $\delta \in (0, 1)$, does the following implementation of DP-Adam satisfy $(\varepsilon/2, \delta)$ -DP (under add/remove neighbouring relation)?

Algorithm 36: DP-Adam

Input: Data $\mathcal{D} = \{x_i\}_{i=1}^N$; epochs E ; batch size B ; learning rate η ; clipping threshold C ; initial parameters $\theta_0 \in \mathbb{R}^p$; initial moments $m_0, v_0 \in \mathbb{R}^p$; decay rates $\beta_1, \beta_2 \in (0, 1)$; small constant $\gamma > 0$.

```

1 Set  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{B}{N}, \lfloor E \cdot N/B \rfloor)$ 
2 for  $t = 0, 1, \dots, \lfloor E \cdot N/B \rfloor - 1$  do
3   Sample a minibatch  $B_t$  via Poisson sampling, where each element of  $\mathcal{D}$  is included
     independently with probability  $B/N$ .
4   for  $x_i \in B_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(x_i)$ 
6      $\tilde{g}_t(x_i) \leftarrow g_t(x_i) \cdot \min(1, \frac{C}{\|g_t(x_i)\|_2})$ 
7    $z_t \sim \mathcal{N}(0, \sigma^2 C^2 I_p)$ 
8    $\bar{g}_t \leftarrow \frac{1}{B} (\sum_{x_i \in B_t} \tilde{g}_t(x_i) + z_t)$ 
9    $m_{t+1} \leftarrow \beta_1 m_t + (1 - \beta_1) \bar{g}_t$ 
10   $v_{t+1} \leftarrow \beta_2 v_t + (1 - \beta_2) \bar{g}_t^{\odot 2}$  //  $\odot$  is elementwise square
11   $\hat{m}_{t+1} \leftarrow \frac{m_{t+1}}{1 - \beta_1^{t+1}}$ 
12   $\hat{v}_{t+1} \leftarrow \frac{v_{t+1}}{1 - \beta_2^{t+1}}$ 
13   $\theta_{t+1} \leftarrow \theta_t - \eta \frac{\hat{m}_{t+1}}{\sqrt{\hat{v}_{t+1} + \gamma}}$ 
14 return  $\theta_{\lfloor E \cdot N/B \rfloor}$ 

```

Label No (False)

Subject DP-ML

Topic dp-adam

Negative Mode conclusion_perturbation

Related Question 100.0

Citation 509

Section appendix A

Publish Year 2021

PDF Link <https://arxiv.org/abs/2110.05679> [Li et al., 2021]

Comments Since the hypothetical oracle $\mathcal{O}$ returns the smallest noise multiplier to guarantee (ε, δ) -DP under the add/remove neighboring relation, it doesn't satisfy the stronger $(\varepsilon/2, \delta)$ -DP.

3.2. DP-GD

Question 26**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship?

Algorithm 37: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
7 return  $\theta_T$ 

```

Label No (False)

Subject DP-ML

Topic DP-GD

Negative Mode atom

Related Question 26

Citation 1206; 736

Section 2;3

Publish Year 2014; 2019

PDF Link <https://arxiv.org/pdf/1405.7085> <https://arxiv.org/pdf/1905.02383> [Bassily et al., 2014]

Comments Since n is private under the add/remove neighboring relation and the given algorithm does not privatize it, the algorithm therefore does not satisfy μ -GDP.

Question 27**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 38: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \max(1, \frac{C}{\|g_i\|_2})$ 
5    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
7 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The gradient clipping is implemented incorrectly: the normalization term should be $\ g_i\ _2$, not $\ g_i\ _2^2$. Thus, the stated privacy guarantee doesn't hold in general.

Question 28**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship?

Algorithm 39: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\bar{g}_t \leftarrow \frac{1}{n} \sum_{i=1}^n \hat{g}_i$ 
6    $\tilde{g}_t \leftarrow \bar{g}_t + \frac{2C\sqrt{T}}{(n-1)\mu} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label Yes (True)

Subject DP-ML

Topic DP-GD

Negative Mode algorithm_perturbation

Related Question 26

Citation 1206; 736

Section 2;3

Publish Year 2014; 2019

PDF Link <https://arxiv.org/pdf/1405.7085> <https://arxiv.org/pdf/1905.02383> [Bassily et al., 2014]

Comments Assume $n > 1$. Under add/remove neighbouring relation, the mean estimator has sensitivity at most $\frac{2C}{n-1}$, thus the noise added in line 6 ensures $\frac{\mu}{\sqrt{T}}$ -GDP per iteration. Since the algorithm runs for a total of T iterations, adaptive composition for Gaussian mechanisms implies that the overall algorithm satisfies μ -GDP (Dong et al. [2022, Corollary 2]).

Question 29**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship?

Algorithm 40: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\bar{g}_t \leftarrow \frac{1}{n} \sum_{i=1}^n \hat{g}_i$ 
6    $\tilde{g}_t \leftarrow \bar{g}_t + \frac{C\sqrt{T}}{(n+1)\mu} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	Under the add/remove neighboring relation, $\bar{g}_t$ can have sensitivity $\frac{2C}{n+1}$. For example, let $X = \{C\}_{i=1}^n$ and $X' = \{C\}_{i=1}^n \cup \{-C\}$. Then $ \text{Avg}(X) - \text{Avg}(X') = \frac{2C}{n+1}$, so the noise scale used in line 6 may be insufficient.

Question 30**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 41: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 Setting noise distribution:  $\xi_0$  is sampled from  $\mathcal{N}(0, I_d)$ , as used in the noise multiplier
  for gaussian mechanism;
2 for  $t = 0, \dots, T-1$  do
3    $\xi_t \sim \mathcal{N}(0, I_d)$ 
4   for  $i = 1, \dots, n$  do
5      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
6      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
7   if  $t \bmod 2 == 0$  then
8      $\xi = \xi_{\lfloor t/2 \rfloor}$ 
9   else
10     $\xi = \xi_t$ 
11    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu} \xi \right)$ 
12    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
13 return  $\theta_T$ 

```

Label No (False)

Subject DP-ML

Topic DP-GD

Negative Mode algorithm_perturbation

Related Question 26

Citation 1206; 736

Section 2;3

Publish Year 2014; 2019

PDF Link <https://arxiv.org/pdf/1405.7085> <https://arxiv.org/pdf/1905.02383> [Bassily et al., 2014]

Comments Line 7-10 reuses noise every two iterations, and therefore does not satisfy the stated privacy guarantee.

Question 31**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship?

Algorithm 42: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; data batches $\mathcal{B} = \{B_1, \dots, B_k\}$, i.e. a partition of D ; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, k$  do
3      $g_i = \frac{1}{|B_i|} \sum_{k \in B_i} \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\tilde{g}_i = g_i + \frac{C\sqrt{T}}{\mu} \xi_i, \quad \xi_i \sim \mathcal{N}(0, I_d)$ 
5      $\tilde{g}_i \leftarrow \tilde{g}_i \cdot \min(1, \frac{C}{\|\tilde{g}_i\|_2})$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \sum_{i=1}^k \tilde{g}_i$ 
7 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The algorithm doesn't satisfy μ -GDP because it swaps the order of clipping and noise addition.

Question 32**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 43: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2    $C = \text{Median}(\nabla_{\theta} \ell(\theta_t; z_1), \dots, \nabla_{\theta} \ell(\theta_t; z_n))$ 
3   for  $i = 1, \dots, n$  do
4      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
5      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
6    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The clipping threshold C is data-dependent and has not been privatized. Therefore, the algorithm does not satisfy the stated privacy guarantee.

Question 33**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 44: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ;
clipping norm $C > 0$; step sizes $\{\eta_t\}_{t=0}^{T-1}$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2    $C' = \text{Median}(\nabla_{\theta} \ell(\theta_t; z_1), \dots, \nabla_{\theta} \ell(\theta_t; z_n)) + \mathcal{N}(0, \frac{4C^2 T}{\mu^2})$ 
3   for  $i = 1, \dots, n$  do
4      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
5      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C'}{\|g_i\|_2})$ 
6    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C' \sqrt{T}}{\mu/2} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The additive noise in line 2 is not calibrated to the sensitivity of the median of the gradients. Therefore, the algorithm does not satisfy the stated privacy guarantee.

Question 34**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 45: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ ; Stopping threshold τ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
   // post-processing
7   if  $\sum_{i=1}^n \ell(\theta_{t+1}; z_i) \geq \tau$  then
8     break
9 return  $\theta_{t+1}$ 

```

Label No (False)

Subject DP-ML

Topic DP-GD

Negative Mode algorithm_perturbation

Related Question 26

Citation 1206; 736

Section 2;3

Publish Year 2014; 2019

PDF Link <https://arxiv.org/pdf/1405.7085> <https://arxiv.org/pdf/1905.02383> [Bassily et al., 2014]

Comments The stopping condition is data-dependent and hasn't been privatized. Therefore, the algorithm does not satisfy the stated privacy guarantee.

Question 35**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 46: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ ; Stopping threshold τ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{2T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
   // running above-threshold
7   if  $\sum_{i=1}^n \ell(\theta_{t+1}; z_i) + \frac{C\sqrt{2T}}{\mu} \xi_t \geq \tau$  then
8     break
9 return  $\theta_{t+1}$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The noise added to the threshold is not properly calibrated to its sensitivity.

Question 36**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 47: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ ; Stopping threshold τ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\tilde{g}_t \leftarrow \frac{1}{n} \left( \sum_{i=1}^n \hat{g}_i + \frac{C\sqrt{2T}}{\mu} \xi_t \right), \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
6    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
7   // running above-threshold
8    $\xi'_t \sim \mathcal{N}(0, I_d)$ 
9   if  $\sum_{i=1}^n \ell(\theta_{t+1}; z_i) + \frac{C\sqrt{2T}}{\mu} \xi'_t \geq \tau$  then
10    break
11 return  $\theta_{t+1}$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	algorithm_perturbation
Related Question	26
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	The noise added to the threshold is not properly calibrated to its sensitivity.

Question 37**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy μ -Gaussian differential privacy under the replace-one neighboring relationship?

Algorithm 48: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\bar{g}_t \leftarrow \frac{1}{n} \sum_{i=1}^n \hat{g}_i$ 
6    $\tilde{g}_t \leftarrow \bar{g}_t + \frac{C\sqrt{T}}{n\mu} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	assumption_perturbation
Related Question	28
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	Since number of sample n is private under the add/remove neighboring relation and the given algorithm does not add enough noise to privatize it, the algorithm therefore does not satisfy μ -GDP.

Question 38**Question:**

The following is an implementation of differentially private gradient descent (DP-GD). Does this implementation satisfy $\mu/2$ -Gaussian differential privacy under the add/remove neighboring relationship (assume number of samples n is public information)?

Algorithm 49: DP-GD with Gaussian Noise

Input : Dataset $D = \{z_i\}_{i=1}^n$; loss $\ell(\theta; z)$; initial parameter $\theta_0 \in \mathbb{R}^d$; iterations T ; step sizes $\{\eta_t\}_{t=0}^{T-1}$; clipping norm $C > 0$; target gaussian DP μ

Output: Final parameter θ_T

```

1 for  $t = 0, \dots, T - 1$  do
2   for  $i = 1, \dots, n$  do
3      $g_i \leftarrow \nabla_{\theta} \ell(\theta_t; z_i)$ 
4      $\hat{g}_i \leftarrow g_i \cdot \min(1, \frac{C}{\|g_i\|_2})$ 
5    $\bar{g}_t \leftarrow \frac{1}{n} \sum_{i=1}^n \hat{g}_i$ 
6    $\tilde{g}_t \leftarrow \bar{g}_t + \frac{C\sqrt{T}}{n\mu} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d)$ 
7    $\theta_{t+1} \leftarrow \theta_t - \eta_t \tilde{g}_t$ 
8 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	DP-GD
Negative Mode	conclusion_perturbation
Related Question	28
Citation	1206; 736
Section	2;3
Publish Year	2014; 2019
PDF Link	https://arxiv.org/pdf/1405.7085 https://arxiv.org/pdf/1905.02383 [Bassily et al., 2014]
Comments	Using the adaptive composition property of Gaussian differential privacy (GDP), the algorithm satisfies μ -GDP, not $\mu/2$ -GDP.

3.3. DP-OLS**Question 39****Question:**

Is the following algorithm (ϵ, δ) -differentially private?

Algorithm 50: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_2 -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Sample $Z \sim \text{Lap}(4B^2/\varepsilon)$ and let $\sigma_{\min}(A)$ denote the smallest singular value of A .

2 **if** $\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$ **then**
 3 Sample a $(r \times n)$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 4 **return** RA and “matrix unaltered”;
 5 **else**
 6 Let A' denote the result of appending A with the $d \times d$ -matrix $wI_{d \times d}$;
 7 Sample a $(r \times (n + d))$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 8 **return** RA' and “matrix altered”;

Label Yes (True)

Subject DP-ML

Topic DP-OLS

Negative Mode None

Related Question None

Citation 167

Section 1

Publish Year 2017

PDF Link <https://proceedings.mlr.press/v70/sheffett17a/sheffett17a.pdf> [Sheffett, 2017]

Comments The given algorithm is Sheffett [2017, Algorithm 1]. The privacy guarantee follows Sheffett [2017, Theorem A.1].

Question 40

Question:

Is the following algorithm (ε, δ) -differentially private?

Algorithm 51: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_∞ -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Sample $Z \sim \text{Lap}(4B^2/\varepsilon)$ and let $\sigma_{\min}(A)$ denote the smallest singular value of A .

2 **if** $\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$ **then**

3 Sample a $(r \times n)$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;

4 **return** RA and “matrix unaltered”;

5 **else**

6 Let A' denote the result of appending A with the $d \times d$ -matrix $wI_{d \times d}$;

7 Sample a $(r \times (n + d))$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;

8 **return** RA' and “matrix altered”;

Label **No (False)**

Subject DP-ML

Topic DP-OLS

Negative Mode assumption_perturbation

Related Question None

Citation 167

Section 1

Publish Year 2017

PDF Link <https://proceedings.mlr.press/v70/sheffett17a/sheffett17a.pdf> [Sheffett, 2017]

Comments The given algorithm is modified from Sheffett [2017, Algorithm 1]. The bound B , originally stated as upper bound for ℓ_2 norm, is changed to bound on ℓ_∞ norm. Since ℓ_2 norm is larger than ℓ_∞ norm, the added noise according to B is not enough.

Question 41

Question:

Is the following algorithm (ε, δ) -differentially private?

Algorithm 52: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_2 -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Let $\sigma_{\min}(A)$ denote the smallest singular value of A .

```

2 if  $\sigma_{\min}(A)^2 > w^2 + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$  then
3   | Sample a  $(r \times n)$ -matrix  $R$  whose entries are i.i.d. samples from a normal Gaussian.;
4   | return  $RA$  and “matrix unaltered”;
5 else
6   | Let  $A'$  denote the result of appending  $A$  with the  $d \times d$ -matrix  $wI_{d \times d}$ ;
7   | Sample a  $(r \times (n + d))$ -matrix  $R$  whose entries are i.i.d. samples from a normal
   | Gaussian.;
8   | return  $RA'$  and “matrix altered”;

```

Label	No (False)
Subject	DP-ML
Topic	DP-OLS
Negative Mode	algorithm_perturbation
Related Question	None
Citation	167
Section	1
Publish Year	2017
PDF Link	https://proceedings.mlr.press/v70/sheff17a/sheff17a.pdf [Sheff17]
Comments	The given algorithm is adapted from Sheff17 [2017, Algorithm 1]. In the PTR test step (line 2), the Laplace noise has been removed. Because this data-dependent step is not privatized, the resulting algorithm is not differentially private.

Question 42

Question:

Is the following algorithm (ε, δ) -differentially private?

Algorithm 53: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_2 -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Sample $Z \sim \text{Lap}(B^2/\varepsilon)$ and let $\sigma_{\min}(A)$ denote the smallest singular value of A .

2 **if** $\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$ **then**
 3 Sample a $(r \times n)$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 4 **return** RA and “matrix unaltered”;
 5 **else**
 6 Let A' denote the result of appending A with the $d \times d$ -matrix $wI_{d \times d}$;
 7 Sample a $(r \times (n + d))$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 8 **return** RA' and “matrix altered”;

Label **No (False)**

Subject DP-ML

Topic DP-OLS

Negative Mode algorithm_perturbation

Related Question None

Citation 167

Section 1

Publish Year 2017

PDF Link <https://proceedings.mlr.press/v70/sheffett17a/sheffett17a.pdf> [Sheffett, 2017]

Comments The given algorithm is adapted from Sheffett [2017, Algorithm 1]. In the PTR test step (line 2), the Laplace noise Z ’s scale has been reduced by a factor of 4, and therefore the stated privacy guarantee no longer holds.

Question 43

Question:

Is the following algorithm $(\varepsilon/2, \delta)$ -differentially private?

Algorithm 54: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_2 -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Sample $Z \sim \text{Lap}(B^2/\varepsilon)$ and let $\sigma_{\min}(A)$ denote the smallest singular value of A .

2 **if** $\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$ **then**
 3 Sample a $(r \times n)$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 4 **return** RA and “matrix unaltered”;
 5 **else**
 6 Let A' denote the result of appending A with the $d \times d$ -matrix $wI_{d \times d}$;
 7 Sample a $(r \times (n + d))$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 8 **return** RA' and “matrix altered”;

Label **No (False)**

Subject DP-ML

Topic DP-OLS

Negative Mode conclusion-perturbation

Related Question 42

Citation 167

Section 1

Publish Year 2017

PDF Link <https://proceedings.mlr.press/v70/sheffett17a/sheffett17a.pdf> [Sheffett, 2017]

Comments The given algorithm is adapted from Sheffett [2017, Algorithm 1]. In the PTR test step (line 2), the Laplace noise Z ’s scale has been reduced by a factor of 4, and therefore the stated privacy guarantee doesn’t hold.

Question 44

Question:

Is the following algorithm (ε, δ) -differentially private, under the add/remove neighbouring relation?

Algorithm 55: Outputting a private Johnson–Lindenstrauss projection of a matrix.

Input: A matrix $A \in \mathbb{R}^{n \times d}$ and a bound $B > 0$ on the ℓ_2 -norm of any row in A ; Privacy parameters: $\varepsilon, \delta > 0$; Parameter r indicating the number of rows in the resulting matrix.

1 Set w such that

$$w^2 = \frac{8B^2}{\varepsilon} \left(\sqrt{2r \ln(8/\delta)} + 2 \ln(8/\delta) \right).$$

Sample $Z \sim \text{Lap}(B^2/\varepsilon)$ and let $\sigma_{\min}(A)$ denote the smallest singular value of A .

2 **if** $\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2 \ln(1/\delta)}{\varepsilon}$ **then**

3 Sample a $(r \times n)$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 4 **return** RA and “matrix unaltered”;

5 **else**

6 Let A' denote the result of appending A with the $d \times d$ -matrix $wI_{d \times d}$;
 7 Sample a $(r \times (n + d))$ -matrix R whose entries are i.i.d. samples from a normal Gaussian.;
 8 **return** RA' and “matrix altered”;

Label **No (False)**

Subject DP-ML

Topic DP-OLS

Negative Mode assumption_perturbation

Related Question 42

Citation 167

Section 1

Publish Year 2017

PDF Link <https://proceedings.mlr.press/v70/sheffett17a/sheffett17a.pdf> [Sheffett, 2017]

Comments (1) The given algorithm is adapted from Sheffett [2017, Algorithm 1]. In the PTR test step (line 2), the Laplace noise Z ’s scale has been reduced by a factor of 4, and therefore the stated privacy guarantee no longer holds;
 (2) under the add/remove neighbouring relation, n is private information.

3.4. dp-sgd

Question 45

Question:

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson subsampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the add/remove neighbouring relation?

Algorithm 56: DP-SGD

Input: loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, iterations T , dataset $D = \{x_1, \dots, x_N\}$, sampling rate q , clipping bound C , learning rate η , initial weights θ_0

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, q, T)$ 
2 for  $t = 0, \dots, T - 1$  do
3    $B \sim \text{PoissonSample}(D, q)$ 
4   for  $x_i \in B$  do
5      $g_i \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_i^{\text{clip}} \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right)$ 
7    $\bar{g} \leftarrow \frac{1}{q} \left( \sum_{i \in B} g_i^{\text{clip}} + \mathcal{N}(0, \sigma^2 C^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \bar{g}$ 
9 return  $\theta_T$ 

```

Label Yes (True)

Subject DP-ML

Topic dp-sgd

Negative Mode None

Related Question 92.0

Citation 0

Section 2

Publish Year 2025

PDF Link <https://arxiv.org/abs/2510.20616> [Rehn et al., 2025]

Comments The example is adopted from Rehn et al. [2025, Alg. 1], using a fixed sampling probability for Poisson sampling.

Question 46**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the poisson sampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the replace-one neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the replace-one neighbouring relation?

Algorithm 57: DP-SGD

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, clipping bound C , initial weights θ_0 , subsample size L

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{L}{N}, T)$ 
2 for  $t = 0, \dots, T-1$  do
3    $L_t \sim \text{PoissonSample}(D, \frac{L}{N})$ 
4   for  $x_i \in L_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_t^{\text{clip}}(x_i) \leftarrow g_t(x_i) \cdot \min\left(1, \frac{C}{\|g_t(x_i)\|_2}\right)$ 
7    $\tilde{g}_t \leftarrow \frac{1}{|L_t|} \left( \sum_{x_i \in L_t} g_t^{\text{clip}}(x_i) + \mathcal{N}(0, 2\sigma^2 C^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{g}_t$ 
9 return  $\theta_T$ 

```

Label	Yes (True)
Subject	DP-ML
Topic	dp-sgd
Negative Mode	None
Related Question	93.0
Citation	9020
Section	3.1
Publish Year	2016
PDF Link	https://dl.acm.org/doi/pdf/10.1145/2976749.2978318 [Abadi et al., 2016]
Comments	This algorithm is adopted from Abadi et al. [2016, Algo. 2]

Question 47**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the poisson sampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the replace-one neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the replace-one neighbouring relation?

Algorithm 58: DP-SGD

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, clipping bound C , initial weights θ_0 , subsample size L

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{L}{N}, T)$ 
2 for  $t = 0, \dots, T-1$  do
3    $L_t \sim \text{PoissonSample}(D, \frac{L}{N})$ 
4   for  $x_i \in L_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_t^{\text{clip}}(x_i) \leftarrow g_t(x_i) \cdot \min\left(\frac{1}{C}, \frac{1}{\|g_t(x_i)\|_2}\right)$ 
7    $\tilde{g}_t \leftarrow \frac{1}{|L_t|} \left( \sum_{x_i \in L_t} g_t^{\text{clip}}(x_i) + \mathcal{N}(0, 2\sigma^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{g}_t$ 
9 return  $\theta_T$ 

```

Label Yes (True)

Subject DP-ML

Topic dp-sgd

Negative Mode None

Related Question 94.0

Citation 281

Section 2

Publish Year 2022

PDF Link <https://arxiv.org/pdf/2204.13650> [De et al., 2022]

Comments Under the replace-one neighboring relation, $\sum_{x_i \in L_t} g_t^{\text{clip}}(x_i)$ has L_2 sensitivity $\sqrt{2}$. Therefore, by the oracle's guarantee, the added noise ensures (ε, δ) -DP.

Question 48**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the poisson sampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the add/remove neighbouring relation?

Algorithm 59: DP-SGD

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, clipping bound C , initial weights θ_0 , subsample size L

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{L}{N}, T)$ 
2 for  $t = 0, \dots, T-1$  do
3    $L_t \sim \text{PoissonSample}(D, \frac{L}{N})$ 
4   for  $x_i \in L_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_t^{\text{clip}}(x_i) \leftarrow g_t(x_i) \cdot \min\left(1, \frac{C}{\|g_t(x_i)\|_2}\right)$ 
7    $\tilde{g}_t \leftarrow \frac{1}{|L_t|} \left( \sum_{x_i \in L_t} g_t^{\text{clip}}(x_i) + \mathcal{N}(0, \sigma^2 C^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{g}_t$ 
9 return  $\theta_T$ 

```

Label No (False)

Subject DP-ML

Topic dp-sgd

Negative Mode assumption_perturbation

Related Question 93.0

Citation 9020

Section 3.1

Publish Year 2016

PDF Link <https://dl.acm.org/doi/pdf/10.1145/2976749.2978318> [Abadi et al., 2016]

Comments Since the number of samples N is private information under the add/remove neighboring relation, the algorithm directly uses N to calculate the subsampling rate for Poisson sampling, thus not private in general.

Question 49**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the poisson sampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy ε -Gaussian DP for some $\varepsilon > 0$?

Algorithm 60: DP-SGD

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, clipping bound C , initial weights θ_0 , subsample size L

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{L}{N}, T)$ 
2 for  $t = 0, \dots, T-1$  do
3    $L_t \sim \text{PoissonSample}(D, \frac{L}{N})$ 
4   for  $x_i \in L_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_t^{\text{clip}}(x_i) \leftarrow g_t(x_i) \cdot \min\left(1, \frac{C}{\|g_t(x_i)\|_2}\right)$ 
7    $\tilde{g}_t \leftarrow \frac{1}{|L_t|} \left( \sum_{x_i \in L_t} g_t^{\text{clip}}(x_i) + \mathcal{N}(0, \sigma^2 C^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{g}_t$ 
9 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	dp-sgd
Negative Mode	conclusion_perturbation
Related Question	93.0
Citation	9020
Section	3.1
Publish Year	2016
PDF Link	https://dl.acm.org/doi/pdf/10.1145/2976749.2978318 [Abadi et al., 2016]
Comments	(1) Since the number of samples N is private information under the add/remove neighboring relation, the algorithm directly uses N to calculate the subsampling rate for Poisson sampling, thus not private in general. (2) Notice that the subsampled Gaussian mechanism does not satisfy pure Gaussian DP. The algorithm is a composition of Gaussian mechanisms with Poisson subsampling, thus doesn't satisfy pure GDP. In conclusion, statement is incorrect.

Question 50**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson sampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the replace-one neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under replace-one neighbouring relation?

Algorithm 61: DP-SGD

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, clipping bound C , initial weights θ_0 , subsample size L

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, \frac{L}{N}, T)$ 
2 for  $t = 0, \dots, T-1$  do
3    $L_t \sim \text{PoissonSample}(D, \frac{L}{N})$ 
4   for  $x_i \in L_t$  do
5      $g_t(x_i) \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_t^{\text{clip}}(x_i) \leftarrow g_t(x_i) \cdot \min(1, C)$ 
7    $\tilde{g}_t \leftarrow \frac{1}{|L_t|} \left( \sum_{x_i \in L_t} g_t^{\text{clip}}(x_i) + \mathcal{N}(0, 2\sigma^2 I_d) \right)$ 
8    $\theta_{t+1} \leftarrow \theta_t - \eta \tilde{g}_t$ 
9 return  $\theta_T$ 

```

Label	No (False)
Subject	DP-ML
Topic	dp-sgd
Negative Mode	algorithm_perturbation
Related Question	93.0
Citation	9020
Section	3.1
Publish Year	2016
PDF Link	https://dl.acm.org/doi/pdf/10.1145/2976749.2978318 [Abadi et al., 2016]
Comments	The provided DP-SGD implementation is incorrect because the noise scale is not calibrated to the sensitivity of the clipped gradients. In standard DP-SGD, the added Gaussian noise should have standard deviation $\sigma \cdot \text{sens}(g^{\text{clip}})$. However, the algorithm instead uses $\sqrt{2}\sigma$, which in general does not satisfy the stated privacy guarantee.

Question 51**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson subsampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the add/remove neighbouring relation?

Algorithm 62: DP-SGD

Input: loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, iterations T , dataset $D = \{x_1, \dots, x_N\}$, sampling rate q , clipping bound C , learning rate η , initial weights θ_0

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, q, T)$ 
2 for  $t = 0, \dots, T - 1$  do
3    $B \sim \text{PoissonSample}(D, q)$ 
4   for  $x_i \in B$  do
5      $g_i \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $\tilde{g}_i \leftarrow g_i + \mathcal{N}(0, \sigma^2 C^2 I_d)$ 
7      $g_i^{\text{clip}} \leftarrow \tilde{g}_i \cdot \min\left(1, \frac{C}{\|\tilde{g}_i\|_2}\right)$ 
8    $\bar{g} \leftarrow \frac{1}{q} \left( \sum_{i \in B} g_i^{\text{clip}} \right)$ 
9    $\theta_{t+1} \leftarrow \theta_t - \eta \bar{g}$ 
10 return  $\theta_T$ 

```

Label No (False)

Subject DP-ML

Topic dp-sgd

Negative Mode algorithm_perturbation

Related Question 92.0

Citation 0

Section 2

Publish Year 2025

PDF Link <https://arxiv.org/abs/2510.20616> [Rehn et al., 2025]

Comments The provided DP-SGD implementation is incorrect because it reverses the order of gradient clipping and noise addition. Since the noise scale is calibrated to the sensitivity of the clipped gradient, adding noise before clipping generally does not guarantee the stated differential privacy.

Question 52**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson subsampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the add/remove neighbouring relation?

Algorithm 63: DP-SGD

Input: loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, iterations T , dataset $D = \{x_1, \dots, x_N\}$, sampling rate q , clipping bound C , initial learning rate η_0 , initial weights θ_0

```

1 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, q, T)$ 
2 for  $t = 0, \dots, T - 1$  do
3    $B \sim \text{PoissonSample}(D, q)$ 
4   for  $x_i \in B$  do
5      $g_i \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
6      $g_i^{\text{clip}} \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right)$ 
7    $\bar{g} \leftarrow \frac{1}{q} \left( \sum_{i \in B} g_i^{\text{clip}} + \mathcal{N}(0, \sigma^2 C^2 I_d) \right)$ 
8   if  $T \geq 10$  and  $\mathcal{L}(\theta_t) > \mathcal{L}(\theta_{t-1})$  then
9      $\eta_t = \eta_{t-1}/2$ 
10  else
11     $\eta_t = \eta_{\max\{0, t-1\}}$ 
12   $\theta_{t+1} \leftarrow \theta_t - \eta_t \bar{g}$ 
13 return  $\theta_T$ 

```

Label No (False)

Subject DP-ML

Topic dp-sgd

Negative Mode algorithm_perturbation

Related Question 92.0

Citation 0

Section 2

Publish Year 2025

PDF Link <https://arxiv.org/abs/2510.20616> [Rehn et al., 2025]

Comments The provided DP-SGD implementation doesn't satisfy DP since the learning-rate scheduler depends on the (data-dependent) loss and is not privatized, so the overall procedure doesn't not satisfy DP in general.

Question 53**Question:**

Given a privacy accountant $\mathcal{O}$ such that, given target privacy parameters (ε, δ) , the Poisson subsampling rate q , and the number of DP-SGD iterations T (i.e., the number of compositions), returns the calibrated noise multiplier σ for the subsampled Gaussian mechanism used in DP-SGD under the add/remove neighbouring relation. Can the following implementation of DP-SGD satisfy (ε, δ) -DP for some $\varepsilon > 0$ and $\delta \in (0, 1)$, under the add/remove neighbouring relation?

Algorithm 64: DP-SGD with Warm-up Initialization

Input: iterations T , dataset $D = \{x_1, \dots, x_N\}$, loss function $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in D} \mathcal{L}(\theta, x)$, sampling rate q , clipping bound C , learning rate η , warm-up learning rate η_w , group size L , publicly pretrained weight ϕ_0

```

1 for  $s = 0, \dots, \lfloor \frac{T}{2} \rfloor - 1$  do
2    $x_s \sim \text{Unif}(D)$ 
3    $g_s^w \leftarrow \nabla_{\phi_s} \mathcal{L}(\phi_s, x_s)$ 
4    $\phi_{s+1} \leftarrow \phi_s - \eta_w g_s^w$ 
5 Set:  $\theta_0 \leftarrow \phi_{\lfloor \frac{T}{2} \rfloor}$ .
6 Set:  $\sigma \leftarrow \mathcal{O}(\varepsilon, \delta, q, T)$ 
7 for  $t = 0, \dots, T - 1$  do
8    $B \sim \text{PoissonSample}(D, q)$ 
9   for  $x_i \in B$  do
10     $g_i \leftarrow \nabla_{\theta_t} \mathcal{L}(\theta_t, x_i)$ 
11     $g_i^{\text{clip}} \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right)$ 
12     $\bar{g} \leftarrow \frac{1}{q} \left( \sum_{i \in B} g_i^{\text{clip}} + \mathcal{N}(0, \sigma^2 C^2 I_d) \right)$ 
13     $\theta_{t+1} \leftarrow \theta_t - \eta \bar{g}$ 
14 return  $\{\theta_t\}_{t=0}^T$ 

```

Label No (False)

Subject DP-ML

Topic dp-sgd

Negative Mode algorithm_perturbation

Related Question 92.0

Citation 0

Section 2

Publish Year 2025

PDF Link <https://arxiv.org/abs/2510.20616> [Rehn et al., 2025]

Comments The provided DP-SGD implementation does not satisfy differential privacy, because its warm-up initialization procedure is data-dependent and not privatized. As a result, the overall algorithm does not satisfy DP in general.

3.5. Hyperparameter-tuning

Question 66

Question:

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1, \infty)$ and $\gamma \in (0, 1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{0, \gamma}$

and each M_j satisfies (α, ε) -RDP for all $\alpha > 1$, the algorithm satisfies (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 65: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score($\cdot$) computed on a public holdout set

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2 for  $k \leftarrow 1, 2, \dots, K$  do
3   Sample  $j_k \sim \text{Unif}(\{1, \dots, m\})$ 
4    $\theta_k \leftarrow M_{j_k}(D)$ 
5    $s_k \leftarrow \text{score}(\theta_k)$ 
6  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
7  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
8 return  $(\hat{\theta}, j^*)$ 

```

Label	Yes (True)
Subject	DP-ML
Topic	Hyperparameter-tuning
Negative Mode	atom
Related Question	66
Citation	159
Section	3.4
Publish Year	2022
PDF Link	https://arxiv.org/pdf/2110.03620 [Papernot and Steinke, 2021]
Comments	This follows from Papernot and Steinke [2021, Theorem 2], instantiated with $\text{TNB}_{0, \gamma}$.

Question 67
Question:

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1, \infty)$ and $\gamma \in (0, 1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{0, \gamma}$ and each M_j satisfies (α, ε) -RDP for all $\alpha > 1$, the algorithm satisfies (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 66: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric
 score : model $\times$ dataset $\rightarrow \mathbb{R}_+$

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2 Split  $D$  into  $D_{train}$  and  $D_{test}$ 
3 for  $k \leftarrow 1, 2, \dots, K$  do
4   Sample  $j_k \sim \text{Unif}(\{1, \dots, m\})$ 
5    $\theta_k \leftarrow M_{j_k}(D_{train})$ 
6    $s_k \leftarrow \text{score}(\theta_k, D_{test})$ 
7  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
8  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
9 return  $(\hat{\theta}, j^*)$ 

```

Label	No (False)
Subject	DP-ML
Topic	Hyperparameter-tuning
Negative Mode	algorithm_perturbation
Related Question	66
Citation	159
Section	3.4
Publish Year	2022
PDF Link	https://arxiv.org/pdf/2110.03620 [Papernot and Steinke, 2021]
Comments	The validation score is computed on the private dataset D_{test} , and the $\arg \max$ is taken directly afterward. Therefore, the algorithm is not differentially private.

Question 68**Question:**

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1, \infty)$ and $\gamma \in (0, 1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{0, \gamma}$ and each M_j satisfies (α, ε) -RDP for all $\alpha > 1$, the algorithm satisfies (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 67: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric
 score : model $\times$ dataset $\rightarrow \mathbb{R}_+$

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2 for  $k \leftarrow 1, 2, \dots, K$  do
3   Sample  $j_k \sim \text{Unif}(\{1, \dots, m\})$ 
4    $\theta_k \leftarrow M_{j_k}(D)$ 
5    $s_k \leftarrow \text{score}(\theta_k, D)$ 
6  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
7  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
8 return  $(\hat{\theta}, j^*)$ 

```

Label No (False)

Subject DP-ML

Topic Hyperparameter-tuning

Negative Mode algorithm_perturbation

Related Question 66

Citation 159

Section 3.4

Publish Year 2022

PDF Link <https://arxiv.org/pdf/2110.03620> [Papernot and Steinke, 2021]

Comments The validation score is computed on the private dataset D , and the $\arg \max$ is taken directly afterward. Therefore, the algorithm is not differentially private.

Question 69**Question:**

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1, \infty)$ and $\gamma \in (0, 1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{0, \gamma}$ and each M_j satisfies (α, ε) -RDP for all $\alpha > 1$, the algorithm satisfies (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 68: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score($\cdot$) computed on a public holdout set

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2  $\mathcal{M} \leftarrow \text{Shuffle}(\{1, 2, 3, \dots, m\})$ 
3 for  $k \leftarrow 1, 2, \dots, K$  do
4    $\text{index}_k = k \bmod m$ 
5    $j_k = \mathcal{M}[\text{index}]$ 
6    $\theta_k \leftarrow M_{j_k}(D)$ 
7    $s_k \leftarrow \text{score}(\theta_k, D)$ 
8  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
9  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
10 return  $(\hat{\theta}, j^*)$ 

```

Label	No (False)
Subject	DP-ML
Topic	Hyperparameter-tuning
Negative Mode	algorithm_perturbation
Related Question	66
Citation	159
Section	3.4
Publish Year	2022
PDF Link	https://arxiv.org/pdf/2110.03620 [Papernot and Steinke, 2021]
Comments	The arg max is taken over all candidates without random stopping. Therefore, the algorithm does not satisfy the stated privacy guarantee.

Question 70**Question:**

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution with parameters $\eta \in (-1, \infty)$ and $\gamma \in (0, 1)$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{0, \gamma}$ and each M_j satisfies (α, ε^2) -RDP with $\varepsilon > 1$ for all $\alpha > 1$, the algorithm satisfies (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(1-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$.

Algorithm 69: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score($\cdot$) computed on a public holdout set

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2 for  $k \leftarrow 1, 2, \dots, K$  do
3   Sample  $j_k \sim \text{Unif}(\{1, \dots, m\})$ 
4    $\theta_k \leftarrow M_{j_k}(D)$ 
5    $s_k \leftarrow \text{score}(\theta_k)$ 
6  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
7  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
8 return  $(\hat{\theta}, j^*)$ 

```

Label No (False)

Subject DP-ML

Topic Hyperparameter-tuning

Negative Mode assumption_perturbation

Related Question 66

Citation 159

Section 3.4

Publish Year 2022

PDF Link <https://arxiv.org/pdf/2110.03620> [Papernot and Steinke, 2021]

Comments Consider the following counterexample. Let $\varepsilon = 4$ and suppose each M_j is a 16-DP mechanism. Taking $\alpha \rightarrow \infty$ yields $\varepsilon' = 8$, which would imply that the overall hyperparameter selection algorithm is 8-DP—i.e., it would provide a stronger privacy guarantee than the underlying mechanisms. This cannot be true.

Question 71**Question:**

Let $\text{TNB}_{\eta, \gamma}$ denote the truncated Negative Binomial distribution: with parameters $\eta \in (-1, \infty)$ and $\gamma > 0$, supported on positive integers. For $\eta \neq 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{\gamma^{-\eta}-1} \prod_{\ell=0}^{k-1} \frac{\ell+\eta}{\ell+1}$ and $\mathbb{E}[K] = \frac{\eta(1-\gamma)}{\gamma(1-\gamma^\eta)}$. For $\eta = 0$, $\Pr[K = k] = \frac{(1-\gamma)^k}{k \log(1/\gamma)}$ and $\mathbb{E}[K] = \frac{1/\gamma-1}{\log(1/\gamma)}$. If $\mathcal{D} = \text{TNB}_{\gamma, 3}$ with $\gamma = 3$ and each M_j satisfies (α, ε) -RDP with $\varepsilon > 1$ for all $\alpha > 1$, does the algorithm satisfy (α, ε') -RDP with $\varepsilon' = (2 - \frac{1}{\alpha})\varepsilon + \frac{\log(1/\gamma)}{\alpha} + \frac{\log(5-\gamma) - \log \gamma - \log(\log(1/\gamma))}{\alpha-1}$?

Algorithm 70: Private Hyperparameter Tuning

Input : Dataset D ; DP base training mechanisms $\{M_j\}_{j=1}^m$ (one per hyperparameter setting j); distribution for number of runs $\mathcal{D}$; validation metric score($\cdot$) computed on a public holdout set

Output: Best model $\hat{\theta}$ and chosen hyperparameter j^*

```

1 Draw  $K \sim \mathcal{D}$ 
2 for  $k \leftarrow 1, 2, \dots, K$  do
3   Sample  $j_k \sim \text{Unif}(\{1, \dots, m\})$ 
4    $\theta_k \leftarrow M_{j_k}(D)$ 
5    $s_k \leftarrow \text{score}(\theta_k)$ 
6  $k^* \leftarrow \arg \max_{k \in \{1, \dots, K\}} s_k$ 
7  $j^* \leftarrow j_{k^*}, \hat{\theta} \leftarrow \theta_{k^*}$ 
8 return  $(\hat{\theta}, j^*)$ 

```

Label No (False)

Subject DP-ML

Topic Hyperparameter-tuning

Negative Mode assumption_perturbation

Related Question 66

Citation 159

Section 3.4

Publish Year 2022

PDF Link <https://arxiv.org/pdf/2110.03620> [Papernot and Steinke, 2021]

Comments To define a valid probability distribution, γ must lie in $[0, 1]$; otherwise the resulting density can take negative values. Therefore, the statement is false.

3.6. objective perturbation

Question 87

Question:

Suppose that for all $z \in \mathcal{Z}$, $\ell(\cdot, z)$ is twice-differentiable, and the rank of its Hessian $\nabla^2 \ell(\mathbf{w}, z)$ at any $\mathbf{w} \in \mathcal{W}$ is at most 1. Also assume that the smoothness parameter satisfies $\beta \leq \epsilon n \lambda$. Is the following algorithm (ϵ, δ) -differentially private?

Algorithm 71: $\mathcal{A}_{\text{ObjP}}$: Objective Perturbation

Input: Private dataset $S = (z_1, \dots, z_n) \in \mathcal{Z}^n$, L -Lipschitz, β -smooth, convex loss function ℓ , convex set $\mathcal{W} \subseteq \mathbb{R}^d$, privacy parameters $\epsilon \leq 1$, $\delta \leq 1/n^2$, regularization parameter λ .

```

1 1: Sample  $\mathbf{G} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ , where  $\sigma^2 = \frac{10L^2 \log(1/\delta)}{\epsilon^2}$ ;
2 2: return  $\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} \hat{\mathcal{L}}(\mathbf{w}; S) + \frac{\langle \mathbf{G}, \mathbf{w} \rangle}{n} + \lambda \|\mathbf{w}\|^2$ , where  $\hat{\mathcal{L}}(\mathbf{w}; S) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{w}, z_i)$ ;

```

Label	Yes (True)
Subject	DP-ML
Topic	objective perturbation
Negative Mode	None
Related Question	None
Citation	528
Section	2.2; 5
Publish Year	2012
PDF Link	https://proceedings.mlr.press/v23/kifer12/kifer12.pdf https://arxiv.org/pdf/1908.09970 [Kifer et al., 2012, Bassily et al., 2019]
Comments	The given algorithm is from Bassily et al. [2019, Algorithm 2]. The privacy guarantee follows Bassily et al. [2019, Theorem 5.2].

Question 88

Question:

Suppose that for all $z \in \mathcal{Z}$, $\ell(\cdot, z)$ is twice-differentiable, and the rank of its Hessian $\nabla^2 \ell(\mathbf{w}, z)$ at any $\mathbf{w} \in \mathcal{W}$ is at most 1. Also assume that the smoothness parameter satisfies $\beta \leq \epsilon n \lambda$. Is the following algorithm (ϵ, δ) -differentially private?

Algorithm 72: $\mathcal{A}_{\text{ObjP}}$: Objective Perturbation

Input: Private dataset $S = (z_1, \dots, z_n) \in \mathcal{Z}^n$, L -Lipschitz, β -smooth, loss function ℓ , privacy parameters $\epsilon \leq 1$, $\delta \leq 1/n^2$, regularization parameter λ .

- 1 **1:** Sample $\mathbf{G} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, where $\sigma^2 = \frac{10L^2 \log(1/\delta)}{\epsilon^2}$;
 - 2 **2:** return $\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} \hat{\mathcal{L}}(\mathbf{w}; S) + \frac{\langle \mathbf{G}, \mathbf{w} \rangle}{n} + \lambda \|\mathbf{w}\|^2$, where $\hat{\mathcal{L}}(\mathbf{w}; S) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{w}, z_i)$;
-

Label	No (False)
Subject	DP-ML
Topic	objective perturbation
Negative Mode	assumption_perturbation
Related Question	None
Citation	528
Section	2.2; 5
Publish Year	2012
PDF Link	https://proceedings.mlr.press/v23/kifer12/kifer12.pdf https://arxiv.org/pdf/1908.09970 [Kifer et al., 2012, Bassily et al., 2019]
Comments	The given algorithm is modified from Bassily et al. [2019, Algorithm 2], removing the convexity assumption on loss function ℓ , thus doesn't satisfy DP in general.

Question 89
Question:

Suppose that for all $z \in \mathcal{Z}$, $\ell(\cdot, z)$ is twice-differentiable, and the rank of its Hessian $\nabla^2 \ell(\mathbf{w}, z)$ at any $\mathbf{w} \in \mathcal{W}$ is at most 1. Also assume that the smoothness parameter satisfies $\beta \leq \epsilon n \lambda$. Is the following algorithm (ϵ, δ) -differentially private?

Algorithm 73: $\mathcal{A}_{\text{ObjP}}$: Objective Perturbation

Input: Private dataset $S = (z_1, \dots, z_n) \in \mathcal{Z}^n$, L -Lipschitz, β -smooth, convex loss function ℓ , convex set $\mathcal{W} \subseteq \mathbb{R}^d$, privacy parameters $\epsilon \leq 1$, $\delta \leq 1/n^2$, regularization parameter λ , gradient solver $\mathcal{O}$

- 1 **1:** Sample $\mathbf{G} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, where $\sigma^2 = \frac{10L^2 \log(1/\delta)}{\epsilon^2}$;
 - 2 **2:** Define $f(\mathbf{w}) = \hat{\mathcal{L}}(\mathbf{w}; S) + \frac{\langle \mathbf{G}, \mathbf{w} \rangle}{n} + \lambda \|\mathbf{w}\|^2$, where $\hat{\mathcal{L}}(\mathbf{w}; S) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{w}, z_i)$;
 - 3 **3:** Specify the number of iterations T and the feasible region $\mathcal{W}$. Apply gradient solver $\mathcal{O}$ to $f(\mathbf{w})$ (constrained to $\mathcal{W}$) and output the solution path $\{\hat{\mathbf{w}}_t\}_{t=1}^T$.
 - 4 **return** $\{\hat{\mathbf{w}}_t\}_{t=1}^T$
-

Label	No (False)
Subject	DP-ML
Topic	objective perturbation
Negative Mode	algorithm_perturbation
Related Question	87
Citation	528
Section	2.2; 5
Publish Year	2012
PDF Link	https://proceedings.mlr.press/v23/kifer12/kifer12.pdf https://arxiv.org/pdf/1908.09970 [Kifer et al., 2012, Bassily et al., 2019]
Comments	Objective perturbation guarantees DP only for the released minimizer. It does not, in general, provide DP for the entire released optimization trajectory.

Question 90**Question:**

Suppose that for all $z \in \mathcal{Z}$, $\ell(\cdot, z)$ is twice-differentiable, and the rank of its Hessian $\nabla^2 \ell(\mathbf{w}, z)$ at any $\mathbf{w} \in \mathcal{W}$ is at most 1. Also assume that the smoothness parameter satisfies $\beta \leq \epsilon n \lambda$. Is the following algorithm $(\epsilon/2, \delta)$ -differentially private?

Algorithm 74: $\mathcal{A}_{\text{ObjP}}$: Objective Perturbation

Input: Private dataset $S = (z_1, \dots, z_n) \in \mathcal{Z}^n$, L -Lipschitz, β -smooth, convex loss function ℓ , convex set $\mathcal{W} \subseteq \mathbb{R}^d$, privacy parameters $\varepsilon \leq 1$, $\delta \leq 1/n^2$, regularization parameter λ .

- 1 **1:** Sample $\mathbf{G} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, where $\sigma^2 = \frac{10L^2 \log(1/\delta)}{\varepsilon^2}$;
- 2 **2:** return $\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} \hat{\mathcal{L}}(\mathbf{w}; S) + \frac{\langle \mathbf{G}, \mathbf{w} \rangle}{n} + \lambda \|\mathbf{w}\|^2$, where $\hat{\mathcal{L}}(\mathbf{w}; S) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{w}, z_i)$;

Label	No (False)
Subject	DP-ML
Topic	objective perturbation
Negative Mode	conclusion_perturbation
Related Question	87
Citation	528
Section	2.2;5
Publish Year	2012
PDF Link	https://proceedings.mlr.press/v23/kifer12/kifer12.pdf https://arxiv.org/pdf/1908.09970 [Kifer et al., 2012, Bassily et al., 2019]
Comments	Take $d = 1$ and $\mathcal{W} = [-1, 1]$. Let the loss be $\ell(w, z) = zw$ with $z \in \{-1, +1\}$. Then $\ell(\cdot, z)$ is convex and twice differentiable, $L = 1$ -Lipschitz since $ \partial_w \ell(w, z) = z \leq 1$, $\beta = 0$ -smooth, and $\nabla^2 \ell(w, z) = 0$ so $\text{rank}(\nabla^2 \ell(w, z)) = 0 \leq 1$; hence $\beta \leq \varepsilon n \lambda$ holds for any ε, n, λ . Fix $n = 100$, $\varepsilon = 1$, $\delta = 1/n^2 = 10^{-4}$, $\lambda = 1$, $L = 1$. Consider replace-one neighboring datasets S and S' where S has all $z_i = +1$ and S' differs in one example with $z_1 = -1$. Then $\hat{\mathcal{L}}(w; S) = w$ and $\hat{\mathcal{L}}(w; S') = 0.98w$. The algorithm outputs $\hat{w}(S) = \arg \min_{w \in [-1, 1]} (w + \frac{G}{n}w + \lambda w^2) = -\frac{1+G/n}{2\lambda}$ and $\hat{w}(S') = \arg \min_{w \in [-1, 1]} (0.98w + \frac{G}{n}w + \lambda w^2) = -\frac{0.98+G/n}{2\lambda}$, where $G \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \frac{10L^2 \log(1/\delta)}{\varepsilon^2}$. Thus $\hat{w}(S)$ and $\hat{w}(S')$ are Gaussians with common standard deviation $s = \frac{\sigma}{2\lambda n}$ and mean shift $\Delta = \mathbb{E}[\hat{w}(S')] - \mathbb{E}[\hat{w}(S)] = \frac{1-0.98}{2\lambda} = 0.01$. With $\delta = 10^{-4}$ and $\varepsilon = 1$, $\sigma = \sqrt{10 \log(10^4)} \approx 9.597$, hence $s = \frac{9.597}{2 \cdot 1 \cdot 100} \approx 0.047985$. For two Gaussians with equal variance, the tight δ at privacy level ε_0 is $\delta(\varepsilon_0) = \Phi(\frac{\Delta}{2s} - \varepsilon_0 \frac{s}{\Delta}) - e^{\varepsilon_0} \Phi(-\frac{\Delta}{2s} - \varepsilon_0 \frac{s}{\Delta})$, where Φ is the standard normal CDF. Plugging $\Delta = 0.01$, $s \approx 0.047985$, and $\varepsilon_0 = \varepsilon/2 = 0.5$ yields $\delta(0.5) \approx 7.26 \times 10^{-4} > 10^{-4}$. Therefore the algorithm is not $(\varepsilon/2, \delta)$ -DP in general.

3.7. output perturbation**Question 91****Question:**

Let $\varepsilon > 0$, $\delta \in [0, \frac{1}{2})$. Let $F : \mathbb{R}^d \times \mathcal{X}^n \rightarrow \mathbb{R}$ be a function with L_2 sensitivity Δ_F and unique minimizer $w^*(X) = \arg \min_{w \in \mathbb{R}^d} F(w, X)$ for all $X \in \mathcal{X}^n$. The (conceptual) output perturbation

algorithms $\mathcal{A}(X) = w_{\mathcal{A}}(X) := w^*(X) + z$ with noise given by density

$$p_z(t) \propto \begin{cases} \exp\left(-\frac{\varepsilon \|t\|_2}{\Delta_F}\right), & \text{if } \delta = 0, \\ \exp\left(-\frac{\varepsilon^2 \|t\|_2^2}{\Delta_F^2 (c_\delta + \sqrt{c_\delta^2 + \varepsilon})^2}\right), & \text{if } \delta \in (0, \frac{1}{2}) \end{cases}$$

where $c_\delta := \sqrt{\log\left(\frac{2}{\sqrt{16\delta^2 + 1} - 1}\right)}$. Is $\mathcal{A}(X)$ (ε, δ) -differentially private?

Label No (False)

Subject DP-ML

Topic output perturbation

Negative Mode None

Related Question None

Citation 2

Section 2.2

Publish Year 2021

PDF Link <https://arxiv.org/pdf/2102.04704> [Lowy and Razaviyayn, 2021]

Comments Not necessarily. Adding Laplace/Gaussian noise to $w^*(X)$ is (ε, δ) -DP only if the noise is calibrated to the L_2 -sensitivity of $w^*(X)$, which does not follow from assuming F has sensitivity Δ_F without additional conditions.

Question 92

Question:

Let $\varepsilon > 0$, $\delta \in [0, \frac{1}{2})$. Let $F : \mathbb{R}^d \times \mathcal{X}^n \rightarrow \mathbb{R}$ be a function with L_2 sensitivity Δ_F and unique minimizer $w^*(X) = \arg \min_{w \in \mathbb{R}^d} F(w, X)$ for all $X \in \mathcal{X}^n$. The (conceptual) output perturbation algorithms $\mathcal{A}(X) = w_{\mathcal{A}}(X) := w^*(X) + z$ with noise given by density

$$p_z(t) \propto \begin{cases} \exp\left(-\frac{\varepsilon \|t\|_2}{\Delta_F}\right), & \text{if } \delta = 0, \\ \exp\left(-\frac{\varepsilon \|t\|_2}{\Delta_F (c_\delta + \sqrt{c_\delta^2 + \varepsilon})}\right), & \text{if } \delta \in (0, \frac{1}{2}) \end{cases}$$

where $c_\delta := \sqrt{\log\left(\frac{2}{\sqrt{16\delta^2 + 1} - 1}\right)}$. Is $\mathcal{A}(X)$ (ε, δ) -differentially private?

Label No (False)

Subject DP-ML

Topic output perturbation

Negative Mode algorithm_perturbation

Related Question	None
Citation	2
Section	2.2
Publish Year	2021
PDF Link	https://arxiv.org/pdf/2102.04704 [Lowy and Razaviyayn, 2021]
Comments	(1) Not necessarily: adding Laplace/Gaussian noise to $w^*(X)$ is (ε, δ) -DP only if the noise is calibrated to the L_2 -sensitivity of $w^*(X)$, which does not follow from assuming F has sensitivity Δ_F without additional stability/regularity conditions; (2) the density for $\delta \in (0, 1/2)$ is not correct.

Question 93

Question:

Let $\varepsilon > 0$, $\delta \in [0, \frac{1}{2})$. Let $F : \mathbb{R}^d \times \mathcal{X}^n \rightarrow \mathbb{R}$ be a function and assume it has unique minimizer $w^*(X) = \arg \min_{w \in \mathbb{R}^d} F(w, X)$ for all $X \in \mathcal{X}^n$. We define with L_2 sensitivity $\Delta_F := \sup_{X, X' \in \mathcal{X}^n, |X \Delta X'| \leq 2} \|w^*(X) - w^*(X')\|_2$. The (conceptual) output perturbation algorithms $\mathcal{A}(X) = w_{\mathcal{A}}(X) := w^*(X) + z$ with noise given by density

$$p_z(t) \propto \begin{cases} \exp\left(-\frac{\varepsilon \|t\|_2}{\Delta_F}\right), & \text{if } \delta = 0, \\ \exp\left(-\frac{\varepsilon^2 \|t\|_2^2}{\Delta_F^2 (c_\delta + \sqrt{c_\delta^2 + \varepsilon})^2}\right), & \text{if } \delta \in (0, \frac{1}{2}) \end{cases}$$

where $c_\delta := \sqrt{\log\left(\frac{2}{\sqrt{16\delta^2 + 1} - 1}\right)}$. Is $\mathcal{A}(X)$ (ε, δ) -differentially private?

Label	Yes (True)
Subject	DP-ML
Topic	output perturbation
Negative Mode	algorithm_perturbation
Related Question	None
Citation	2
Section	2.2
Publish Year	2021
PDF Link	https://arxiv.org/pdf/2102.04704 [Lowy and Razaviyayn, 2021]
Comments	This is Lowy and Razaviyayn [2021, proposition 2.1]. The proof is in Lowy and Razaviyayn [2021, Appendix B.1.1].

3.8. PATE

Question 108

Question:

Suppose $\varepsilon(\lambda)$ is defined:

$$\varepsilon(\lambda) := \frac{1}{\lambda - 1} \log \left[(1 - q) \cdot \mathbf{A}(q, \mu_2, \varepsilon_2)^{\lambda-1} + q \cdot \mathbf{B}(q, \mu_1, \varepsilon_1)^{\lambda-1} \right],$$

where $q = \min_{i^* \in [K]} \frac{1}{2} \sum_{i \neq i^*} \text{erfc} \left(\frac{n_{i^*} - n_i}{2\sigma_2} \right)$, $\mu_2 = \sigma_2 \cdot \sqrt{\log(1/q)}$, $\mu_1 = \mu_2 + 1$, $\varepsilon_1 = \mu_1/\sigma_2^2$, $\varepsilon_2 = \mu_2/\sigma_2^2$, $\mathbf{A}(q, \mu_2, \varepsilon_2) = (1 - q) / \left(1 - (qe^{\varepsilon_2})^{\frac{\mu_2-1}{\mu_2}} \right)$ and $\mathbf{B}(q, \mu_1, \varepsilon_1) = e^{\varepsilon_1} / q^{\frac{1}{\mu_1-1}}$. When $q < 1, \mu_1 \geq \lambda, \mu_2 > 1$, and $q \leq e^{(\mu_2-1)\varepsilon_2} / \left(\frac{\mu_1}{\mu_1-1} \cdot \frac{\mu_2}{\mu_2-1} \right)^{\mu_2}$, does the following algorithm satisfy the data dependent $(\lambda, \lambda/\sigma_1^2 + \varepsilon(\lambda))$ -RDP?

Algorithm 75: Confident-GNMax Aggregator

Input : Input query x , threshold T , noise parameters σ_1, σ_2 , K classes

- 1 Compute vote counts $n_j(x)$ for each class $j \in [K]$;
- 2 **if** $\max_{j \in [K]} n_j(x) + \mathcal{N}(0, \sigma_1^2) \geq T$ **then**
- 3 **return** $\arg \max_{j \in [K]} \{n_j(x) + \mathcal{N}(0, \sigma_2^2)\}$
- 4 **else**
- 5 **return** $\perp$

Label Yes (True)

Subject DP-ML

Topic PATE

Negative Mode None

Related Question None

Citation 869

Section 4

Publish Year 2018

PDF Link <https://arxiv.org/pdf/1802.08908> [Papernot et al., 2018]

Comments The proof is by Papernot et al. [2018, Proposition 8, Theorem 6] and sequential composition of Rényi DP [Mironov, 2017]

Question 109**Question:**

Suppose $\varepsilon(\lambda)$ is defined:

$$\varepsilon(\lambda) := \frac{1}{\lambda} \log \left[(1 - q) \cdot \mathbf{A}(q, \mu_2, \varepsilon_2)^{\lambda-1} + q \cdot \mathbf{B}(q, \mu_1, \varepsilon_1)^{\lambda-1} \right],$$

where $q = \min_{i^* \in [K]} \frac{1}{2} \sum_{i \neq i^*} \text{erfc} \left(\frac{n_{i^*} - n_i}{2\sigma_2} \right)$, $\mu_2 = \sigma_2 \cdot \sqrt{\log(1/q)}$, $\mu_1 = \mu_2 + 1$, $\varepsilon_1 = \mu_1/\sigma_2^2$, $\varepsilon_2 = \mu_2/\sigma_2^2$, $\mathbf{A}(q, \mu_2, \varepsilon_2) = (1 - q) / \left(1 - (qe^{\varepsilon_2})^{\frac{\mu_2-1}{\mu_2}} \right)$ and $\mathbf{B}(q, \mu_1, \varepsilon_1) = e^{\varepsilon_1} / q^{\frac{1}{\mu_1-1}}$. When $q < 1, \mu_1 \geq \lambda, \mu_2 > 1$, and $q \leq e^{(\mu_2-1)\varepsilon_2} / \left(\frac{\mu_1}{\mu_1-1} \cdot \frac{\mu_2}{\mu_2-1} \right)^{\mu_2}$, does the following algorithm satisfy the data dependent $(\lambda, \lambda/\sigma_1^2 + \varepsilon(\lambda))$ -RDP?

Algorithm 76: Confident-GNMax Aggregator**Input** : Input query x , threshold T , noise parameters σ_1, σ_2 , K classes

- 1 Compute vote counts $n_j(x)$ for each class $j \in [K]$;
- 2 **if** $\max_{j \in [K]} n_j(x) + \mathcal{N}(0, \sigma_1^2) \geq T$ **then**
- 3 **return** $\arg \max_{j \in [K]} \{n_j(x) + \mathcal{N}(0, \sigma_2^2)\}$
- 4 **else**
- 5 **return** $\perp$

Label No (False)**Subject** DP-ML**Topic** PATE**Negative Mode** conclusion_perturbation**Related Question** None**Citation** 869**Section** 4**Publish Year** 2018**PDF Link** <https://arxiv.org/pdf/1802.08908> [Papernot et al., 2018]

Comments The algorithm is Papernot et al. [2018, Algorithm 1]. The proof is by Papernot et al. [2018, Proposition 8, Theorem 6] and sequential composition of Rényi DP [Mironov, 2017]. The leading factor in $\varepsilon(\lambda)$ is changed from $\frac{1}{\lambda-1}$ to $\frac{1}{\lambda}$, thus the original privacy proof no longer applies and the stated privacy guarantee does not hold in general.

Question 110**Question:**Suppose $\varepsilon(\lambda)$ is defined:

$$\varepsilon(\lambda) := \frac{1}{\lambda-1} \log \left[(1-q) \cdot \mathbf{A}(q, \mu_2, \varepsilon_2)^{\lambda-1} + q \cdot \mathbf{B}(q, \mu_1, \varepsilon_1)^{\lambda-1} \right],$$

where $q = \min_{i^* \in [K]} \frac{1}{2} \sum_{i \neq i^*} \text{erfc} \left(\frac{n_{i^*} - n_i}{2\sigma_2} \right)$, $\mu_2 = \sigma_2 \cdot \sqrt{\log(1/q)}$, $\mu_1 = \mu_2 + 1$, $\varepsilon_1 = \mu_1/\sigma_2^2$, $\varepsilon_2 = \mu_2/\sigma_2^2$, $\mathbf{A}(q, \mu_2, \varepsilon_2) = (1-q) / \left(1 - (qe^{\varepsilon_2})^{\frac{\mu_2-1}{\mu_2}} \right)$ and $\mathbf{B}(q, \mu_1, \varepsilon_1) = e^{\varepsilon_1}/q^{\frac{1}{\mu_1-1}}$. When $q \leq e^{(\mu_2-1)\varepsilon_2} / \left(\frac{\mu_1}{\mu_1-1} \cdot \frac{\mu_2}{\mu_2-1} \right)^{\mu_2}$, does the following algorithm satisfy the data dependent $(\lambda, \lambda/\sigma_1^2 + \varepsilon(\lambda))$ -RDP?

Algorithm 77: Confident-GNMax Aggregator**Input** : Input query x , threshold T , noise parameters σ_1, σ_2 , K classes

- 1 Compute vote counts $n_j(x)$ for each class $j \in [K]$;
- 2 **if** $\max_{j \in [K]} n_j(x) + \mathcal{N}(0, \sigma_1^2) \geq T$ **then**
- 3 **return** $\arg \max_{j \in [K]} \{n_j(x) + \mathcal{N}(0, \sigma_2^2)\}$
- 4 **else**
- 5 **return** $\perp$

Label	No (False)
Subject	DP-ML
Topic	PATE
Negative Mode	assumption_perturbation
Related Question	None
Citation	869
Section	4
Publish Year	2018
PDF Link	https://arxiv.org/pdf/1802.08908 [Papernot et al., 2018]
Comments	The algorithm is Papernot et al. [2018, Algorithm 1]. Since we drop the assumptions $q < 1$, $\mu_1 \geq \lambda$, and $\mu_2 > 1$, the original privacy proof no longer applies and the stated privacy guarantee does not hold in general.

4. Data-Adaptive

4.1. PTR

Question 111

Question:

Given dataset $X \in \mathcal{X}^*$, and query $f : \mathcal{X}^* \rightarrow \mathbb{R}$, we define a proxy:

$$D_{\beta}^f(X) = \min_{X''} \left\{ \text{Hamming}(X, X'') : \max_{X \text{ and } X' \text{ neighbours}} \|f(X) - f(X')\|_1 > \beta \right\}$$

.

Does the following algorithm satisfy (ε, δ) -DP?

Algorithm 78: Propose-Test-Release

Input: Dataset X ; privacy parameters ε, δ ; proposed bound β ; query function $f : \mathcal{X} \rightarrow \mathbb{R}$.

```

1 if  $D_{\beta}^f(X) + \text{Lap}\left(\frac{2}{\varepsilon}\right) \leq \frac{2 \log(1/\delta)}{\varepsilon}$  then
2   | output  $\perp$ 
3 else
4   | release  $f(X) + \text{Lap}\left(\frac{2\beta}{\varepsilon}\right)$ 
```

Label	Yes (True)
Subject	Data-Adaptive
Topic	PTR
Negative Mode	None
Related Question	None
Citation	1104
Section	None

Publish Year	2009
PDF Link	https://www.stat.cmu.edu/~jinglei/dl09.pdf [Dwork and Lei, 2009]
Comments	The algorithm is propose-test-release from Dwork and Lei [2009]. The privacy guarantee follows Vadhan [2017, Proposition 7.3.2]

Question 112

Question:

Given dataset $X \in \mathcal{X}^*$, and query $f : \mathcal{X}^* \rightarrow \mathbb{R}$, we define a proxy:

$$D_{\beta}^f(X) = \min_{X''} \left\{ \text{Hamming}(X, X'') : \max_{X \text{ and } X' \text{ neighbours}} \|f(X) - f(X')\|_1 > \beta \right\}$$

.

Does the following algorithm satisfy (ε, δ) -DP?

Algorithm 79: Propose-Test-Release

Input: Dataset X ; privacy parameters ε, δ ; proposed bound β ; query function $f : \mathcal{X} \rightarrow \mathbb{R}$.

```

1 if  $D_{\beta}^f(X) + \text{Lap}\left(\frac{2}{\varepsilon}\right) \leq \frac{2\log(1/\delta)}{\varepsilon}$  then
2   | output  $\perp$ 
3 else
4   | release  $f(X) + \text{Lap}\left(\frac{\beta}{\varepsilon}\right)$ 
```

Label	No (False)
Subject	Data-Adaptive
Topic	PTR
Negative Mode	algorithm_perturbation
Related Question	None
Citation	1104
Section	None
Publish Year	2009
PDF Link	https://www.stat.cmu.edu/~jinglei/dl09.pdf [Dwork and Lei, 2009]
Comments	The algorithm is modified from propose-test-release from Dwork and Lei [2009]. The release step follows ε -DP when the local sensitivity is upper bounded by β . Thus, following the composition style analysis (Vadhan [2017, Proposition 7.3.2]), the privacy guarantee is $(\frac{3\varepsilon}{2}, \delta)$ -DP, which is weaker than (ε, δ) -DP.

Question 113

Question:

Given a dataset $X \in \mathcal{X}^*$, a proposed noise scale $\Delta > 0$, and a query function $f : \mathcal{X} \rightarrow \mathbb{R}$, define the local sensitivity of f at X as $\Delta_{\text{LS}}(X) := \max_{X'} \left\{ \|f(X) - f(X')\|_1 \mid X' \text{ is a neighbor of } X \right\}$. Suppose we have an oracle $\mathcal{T} : \mathcal{X} \rightarrow \{0, 1\}$ such that: (1) $\mathcal{T}$ satisfies (ε, δ) -differential privacy;

(2) whenever $\Delta_{\text{LS}}(X) \geq \Delta/\varepsilon$, it holds that $\Pr[\mathcal{T}(X) = 1] \leq \delta$. Does the following algorithm satisfy $(2\varepsilon, 2\delta)$ -differential privacy?

Algorithm 80: Propose-test-release-variant

Input: Dataset X ; budget ε ; (ε, δ) -DP oracle $\mathcal{T}$ with false positive rate $\leq \delta$

```

1 if not  $\mathcal{T}(X)$  then
2   output  $\perp$ 
3 else
4   release  $\theta = f(X) + \text{Laplace}(\Delta)$ 

```

Label	Yes (True)
Subject	Data-Adaptive
Topic	PTR
Negative Mode	None
Related Question	None
Citation	10
Section	None
Publish Year	2023
PDF Link	https://arxiv.org/abs/2301.00301 [Redberg et al., 2023]
Comments	The proof follows by Redberg et al. [2023, Theorem 4.3]

Question 114

Question:

Given a dataset $X \in \mathcal{X}^*$, a proposed noise scale $\Delta > 0$, and a query function $f : \mathcal{X} \rightarrow \mathbb{R}$, define the local sensitivity of f at X as $\Delta_{\text{LS}}(X) := \max_{X'} \{ \|f(X) - f(X')\|_1 \mid X' \text{ is a neighbor of } X \}$. Suppose we have an oracle $\mathcal{T} : \mathcal{X} \rightarrow \{0, 1\}$ such that: (1) $\mathcal{T}$ satisfies (ε, δ) -differential privacy; (2) whenever $\Delta_{\text{LS}}(X) \geq \Delta/\varepsilon$, it holds that $\Pr[\mathcal{T}(X) = 1] \leq \delta$. Does the following algorithm satisfy $(2\varepsilon, 2\delta)$ -differential privacy?

Algorithm 81: Propose-test-release-variant

Input: Dataset X ; budget ε ; (ε, δ) -DP oracle $\mathcal{T}$ with false positive rate $\leq \delta$

```

1 if not  $\mathcal{T}(X)$  then
2   output  $\perp$ 
3 else
4   release  $\theta = f(X) + \text{Laplace}(\frac{\Delta_{\text{LS}}}{\varepsilon})$ 

```

Label	No (False)
Subject	Data-Adaptive
Topic	PTR
Negative Mode	algorithm_perturbation
Related Question	None

Citation	10
Section	None
Publish Year	2023
PDF Link	https://arxiv.org/abs/2301.00301 [Redberg et al., 2023]
Comments	This is modified from Redberg et al. [2023, Algorithm 2]. The "release" step is not ε -DP as it add noise calibrated to the local sensitivity, rather than the propose noise scale Δ . Thus the whole procedure is not $(2\varepsilon, 2\delta)$ -DP in general.

4.2. Smooth_sensitivity

Question 127

Question:

Does the following algorithm satisfy 8-zCDP under the replace-one neighbouring relation?

Algorithm 82: Trimmed Mean estimation

Input : Input scalar dataset $D := \{x_1, \dots, x_n\} \in [0, 1]^n$, integer $m \in [0, n/2)$

- 1 Denote the i -th order statistics of D by $x_{(i)}$, where $x_{(1)} \leq \dots \leq x_{(n)}$
 - 2 Let $T = \Pi_{[0,1]} \left(\frac{\sum_{i=m+1}^{n-m} x_{(i)}}{n-2m} \right)$, where Π denotes projection operator
 - 3 Sample $a \sim \mathcal{N}(0, 1)$
 - 4 Calculate $b = \frac{e^a - e^{-a}}{2}$
 - 5 Get scale $c = \max\left\{\frac{x_{(n)} - x_{(m)}}{n-2m}, \exp(-m)\right\}$
 - 6 **return** $T + cb$
-

Label	Yes (True)
Subject	Data-Adaptive
Topic	Smooth_sensitivity
Negative Mode	None
Related Question	None
Citation	105
Section	None
Publish Year	2019
PDF Link	https://arxiv.org/pdf/1906.02830 [Bun and Steinke, 2019]
Comments	The proof follows directly from Bun and Steinke [2019, Propositions 5 and Lemma 50] with $t = s = 1$.

Question 128

Question:

Does the following algorithm satisfy 8-zCDP under the replace-one neighbouring relation?

Algorithm 83: Trimmed Mean estimation**Input** : Input scalar dataset $D := \{x_1, \dots, x_n\} \in [0, 1]^n$, integer $m \in [0, n/2]$

- 1 Denote the i -th order statistics of D by $x_{(i)}$, where $x_{(1)} \leq \dots \leq x_{(n)}$
- 2 Let $T = \Pi_{[0,1]} \left(\frac{\sum_{i=m+1}^{n-m} x_{(i)}}{n-2m} \right)$, where Π denotes projection operator
- 3 Sample $a \sim \mathcal{N}(0, 1)$
- 4 Get scale $c = \max\left\{\frac{x_{(n)} - x_{(m)}}{n-2m}, \exp(-m)\right\}$
- 5 **return** $T + ca$

Label No (False)**Subject** Data-Adaptive**Topic** Smooth_sensitivity**Negative Mode** algorithm_perturbation**Related Question** None**Citation** 105**Section** None**Publish Year** 2019**PDF Link** <https://arxiv.org/pdf/1906.02830> [Bun and Steinke, 2019]

Comments Take $n = 3$ and $m = 1$. Then $T = x_{(2)}$ and $c = \max\{x_{(3)} - x_{(1)}, e^{-1}\}$. Consider replace-one neighbors $D = (0, 0, 1)$ and $D' = (0, 0, 0)$. For D , $T = 0$ and $c = 1$, hence $\mathcal{M}(D) \sim \mathcal{N}(0, 1)$. For D' , $T = 0$ and $c = e^{-1}$, hence $\mathcal{M}(D') \sim \mathcal{N}(0, e^{-2})$. For $P = \mathcal{N}(0, \sigma_1^2)$ and $Q = \mathcal{N}(0, \sigma_2^2)$ with $\sigma_1^2 > \sigma_2^2$, one has $D_\alpha(P \| Q) < \infty$ iff $\alpha < \sigma_1^2 / (\sigma_1^2 - \sigma_2^2)$. Plugging in $\sigma_1^2 = 1$ and $\sigma_2^2 = e^{-2}$ yields the threshold $\alpha \geq 1/(1 - e^{-2}) \approx 1.157$, so in particular $D_2(\mathcal{M}(D) \| \mathcal{M}(D')) = \infty$. Therefore the mechanism is not ρ -zCDP for any finite ρ (hence not 8-zCDP).

Question 129**Question:**

Does the following algorithm satisfy 2-zCDP under the replace-one neighbouring relation?

Algorithm 84: Trimmed Mean estimation**Input** : Input scalar dataset $D := \{x_1, \dots, x_n\} \in [0, 1]^n$, integer $m \in [0, n/2]$

- 1 Denote the i -th order statistics of D by $x_{(i)}$, where $x_{(1)} \leq \dots \leq x_{(n)}$
- 2 Let $T = \Pi_{[0,1]} \left(\frac{\sum_{i=m+1}^{n-m} x_{(i)}}{n-2m} \right)$, where Π denotes projection operator
- 3 Sample $a \sim \text{Laplace}(0, 1)$
- 4 Sample $b \sim \mathcal{N}(0, 1)$
- 5 Get scale $c = \max\left\{\frac{x_{(n)} - x_{(m)}}{n-2m}, \exp(-m)\right\}$
- 6 **return** $T + e^{1.5+b}ac$

Label Yes (True)**Subject** Data-Adaptive

Topic	Smooth_sensitivity
Negative Mode	None
Related Question	None
Citation	105
Section	None
Publish Year	2019
PDF Link	https://arxiv.org/pdf/1906.02830 [Bun and Steinke, 2019]
Comments	The proof follows directly from Bun and Steinke [2019, Propositions 3 and Lemma 50] with $t = \sigma = 1$, $s = \exp(-3/2)$.

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